Resolution refinement followed by the rodeo algorithm for eigenstate preparation: implementation and assessment on the pairing Hamiltonian

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Projective spectral filters such as the rodeo algorithm prepare eigenstates with exponential accuracy in the number of filter cycles, but their cost is set by the squared overlap p of the input state with the target. We implement and assess, on the exactly solvable constant-pairing Hamiltonian, a pipeline in which *resolution refinement* — the adiabatic transport of an eigenstate from a coarse to a fine single-particle basis along a shift-controlled, gapped path — supplies the input of the rodeo filter, and we compare it against the simplest alternatives on the same footing: the same filter applied to the unrefined prolonged state, to an ordinary adiabatic ramp from the Hartree–Fock Hamiltonian, to a prolonged coarse variational state and to full-space unitary coupled-cluster (UCCSD) states. All comparisons use the ground-state infidelity as the common observable, the same second-order product-formula evolution, and a two-qubit gate budget per attempt and per accepted preparation that includes the embedded coarse-Hamiltonian term of the refinement path. For four particles on four levels (eight system qubits) the refined input ($p = 0.998$) is filtered to an infidelity of 5×10^{-4} after two cycles and 1.5×10^{-4} after six, the floor of the averaged post-selection map of the Trotterized filter; the plain adiabatic ramp reaches the same overlap at a quarter of the refinement cost; and the seniority-zero reference admits a four-parameter pair-rotation preparation of 192 CNOTs whose infidelity, 1.8×10^{-4} (energy error 1.4×10^{-3}), is below every target that the filtered pipeline reaches, at two orders of magnitude less cost. Refinement therefore does not beat a cheap, accurate variational input at this system size, nor generic adiabatic preconditioning; its merits are the absence of a classical optimization loop and of training measurements. The extracted refinement time changes with the shift-controlled path gap and with the pairing strength, but the scans are sensitivity studies that identify no scaling exponent, because the response norm that the inverse-gap argument requires to be bounded is itself varied by the shift; the sufficient adiabatic-time and Trotter-step conditions available for such paths are evaluated and found to overestimate the empirical cost by two to three orders of magnitude. Level occupation numbers and correlation energies of the prepared states are reported, and all stages are compiled to rotation–CNOT circuits with gate counts verified against exact evolution. The seniority-conserving pairing force is the cleanest possible testbed; the extension of the assessment to a seniority-breaking particle-hole interaction is the subject of a companion paper.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, and their prospects for nuclear structure have been mapped in recent community studies [13, 52, 53], with schematic nuclear models — pairing, Lipkin, and shell-model Hamiltonians — serving as the standard proving ground [38–40, 42–45]. The state-of-the-art preparation methods, however, each face an obstacle at scale. The variational quantum eigensolver (VQE) [1] is limited by the cost of its classical optimization loop, by the accuracy floor of its ansatz and, for sufficiently expressive circuits, by barren plateaus — gradients that vanish exponentially with system size [2, 51]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [3]. Projective spectral filters — quantum phase estimation [4], polynomial filters based on quantum signal processing [49, 50], projection by measurement [41], and the rodeo algorithm (RA) [5, 6, 46–48] — converge exponentially in the number of cycles, but only once the initial

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state has a significant squared overlap p with the target eigenstate; their cost grows as $1/p$, and for a fixed, unoptimized reference state p falls exponentially with system size.

These limitations are complementary, and the idea of *preconditioning* the input of a projective filter by an adiabatic stage is as old as the rodeo algorithm itself: the original proposal [5] already combines a short adiabatic ramp with the filter, the hierarchical fusion method of Ref. [8] builds large systems from exactly solvable subsystems by alternating adiabatic ramps and rodeo cycles, and the resolution-refinement proposal [7] explicitly recommends switching from adiabatic evolution to a filter such as the rodeo algorithm once the overlap is large. *Resolution refinement* bootstraps an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement changes the low-energy spectrum only mildly, the interpolation path can be kept gapped, and under the additional assumption that the first-order perturbative correction to the tracked state stays bounded [7] the required adiabatic time scales as the inverse gap times a slowly growing system-size factor rather than as the inverse gap squared. The same bootstrapping idea applied along the other axis is *nucleation*, proposed recently by some of us [9]: the exact ground state of a small lattice (the seed) is compiled into a circuit, and the lattice is then grown in stages, each stage a Trotterized adiabatic interpolation that couples newly added qubits to the existing bulk, until the target size is reached. Refinement keeps the domain fixed and raises the resolution; nucleation keeps the resolution fixed and grows the domain. Both replace one hard global preparation by a sequence of local, gapped adiabatic steps that start from an exactly solved problem, both are compiled as Trotterized adiabatic circuits, and Ref. [9] supplies for such circuits sufficient conditions on the evolution time and on the number of Trotter steps, together with a variational variant that reuses the Trotterized circuit as a warm-started ansatz. The present work implements this chain — (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine for a time T; (iii) apply the rodeo algorithm — for a hierarchy of fermionic single-particle bases, and asks the question that decides whether the refinement stage earns its gates: does refinement followed by filtering beat filtering the best input that is available cheaply?

The pairing problem has become a standard target for quantum algorithms, and the programme of Lacroix and collaborators is the closest in spirit to ours. Its common thread is the deliberate use of symmetry breaking and restoration: a symmetry-breaking (BCS-like) trial state is cheap and correlated, and the symmetry is restored afterwards by a projective step — quantum phase estimation used as a projector [23], Grover-type oracles [28], spin filters [25] or classical shadows [31] — with the restored state then handed to a spectral post-processing stage such as phase estimation or a Krylov expansion [24, 26]; Refs. [27, 45] review this viewpoint and Ref. [29] the broader many-body landscape. More recently the same group has used ADAPT-VQE on pairing Hamiltonians with symmetry-conserving and symmetry-breaking operator pools [44], extracted excited states from the subspace spanned along the ADAPT convergence path [32], obtained Lipkin spectra with a quantum equation-of-motion approach [39] and with a two-qubit generator-coordinate scheme [30], and built one-body Green's functions of small superfluid systems from a variational even-N ground state and a quantum subspace expansion for its $N\pm1$ neighbors [33]. Our pipeline addresses the step that all of these share — obtaining a high-fidelity eigenstate of the fine-resolution Hamiltonian from a cheap approximate one — with a different mechanism: not symmetry projection or subspace diagonalization, but an adiabatic path across resolutions followed by an energy filter; the relation is taken up in Sec. VI.

In this work we realize the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [22] and integrable by the Richardson ansatz [35–37]. Its exact solvability is precisely what makes it the right laboratory: every stage of the pipeline can be validated quantitatively — the gate-level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and the rodeo filter against the analytic success-probability formulas of Ref. [6]. All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the demonstration doubles as a reference implementation for hardware.

Relative to the resolution-refinement proposal of Ref. [7], the rodeo literature [5, 6] and the fusion method [8], which established adiabatic preconditioning of the filter on lattice and spin systems, the present work contributes five elements: (i) an implementation of refinement for a hierarchy of *single-particle bases* in a fermionic pairing problem, where the prolongation is gate-free and the embedded coarse Hamiltonian is a conditioned operator whose circuit we give and count; (ii) a quantitative benchmark of the refinement-plus-filter chain against the same filter applied to the unrefined input, to an ordinary adiabatic ramp, to a prolonged coarse variational state and to full-space variational states — the UCCSD template and the four-parameter pair-rotation ansatz that reproduces it exactly — with the ground-state infidelity as the common observable, matched evolution accuracy, and the minimum cost at a given target accuracy, filter cycles included or not; (iii) a gate-level compilation of every stage — Trotterized refinement and Trotterized controlled evolution included — with a two-qubit-gate budget per accepted preparation that separates one-time training, repeated preparation and the filter; (iv) an evaluation of the rigorous sufficient conditions on the adia-

batic time and on the Trotter step count of Ref. [9] for the refinement path, against the resources actually needed; and (v) a reproducible reference implementation in which every figure and number derives from a single executable notebook. The outcome of (ii) is stated plainly: at the system sizes accessible to exact validation here, refinement followed by filtering does not outperform filtering a variational input, is roughly tied with filtering the unrefined prolonged state at matched fidelity, and costs four times more than an ordinary adiabatic ramp from the Hartree–Fock Hamiltonian that reaches the same overlap; the case for the refinement stage rests on its freedom from classical optimization and training measurements, and on regimes — larger spaces, poorer variational inputs — that lie beyond what we can validate exactly.

The pairing force conserves seniority, and that symmetry is what keeps the model transparent: the ground state lives in the seniority-zero subspace, the one-particle–one-hole channel is closed, and single excitations drop out of the variational ansatz. We keep to this seniority-conserving model throughout the present paper, as does most of the quantum-algorithm literature on pairing cited above. In a companion paper [10] we remove the simplification by adding a seniority-breaking particle-hole interaction and repeat the complete assessment — exact and variational references, refinement, filtering and the circuit-level budget — on the resulting Hamiltonian, which carries the one-particle–one-hole channel of realistic two-body interactions.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a^\dagger_{p\uparrow} a^\dagger_{p\downarrow} a_{q\downarrow} a_{q\uparrow}, \tag{1}$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g. The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace, and the model is exactly solvable by the Richardson ansatz [35–37] — the property that anchors every benchmark below. We map level p and spin σ to qubit $2p + \sigma$ (so a k-level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\mathrm{HF}}(N{=}4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed-N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. "Basis refinement" here means *increasing k at fixed N*: the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD) [14, 18], Trotterized into a product of excitation circuits acting on the HF reference,

$$|\Psi(\boldsymbol{\theta})\rangle = \prod_{\mu} e^{\theta_\mu (\hat{\tau}_\mu - \hat{\tau}^\dagger_\mu)} |\mathrm{HF}\rangle \,. \tag{2}$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings, so every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight (the circuits are shown in Fig. 11). The ordered product differs in general from the exponential of the summed cluster operator, so the ansatz is the *factorized* (disentangled) UCCSD, and the operator order — singles first, then doubles, each in the lexicographic order of the pool — is part of its definition. We optimize with the exact analytic gradient of the product ansatz: a fermionic excitation generator has eigenvalues 0 and $\pm i$, so the dependence of the energy on one parameter contains first *and* second harmonics, and the single-frequency two-term parameter-shift rule is not sufficient in general. Because the pairing force does not couple to seniority-changing single excitations, their energy gradient vanishes identically on the seniority-zero manifold; in the zero-initialized optimizations used here all singles amplitudes stay at zero to machine precision and the solver is effectively a *doubles* theory. (This is a property of the optimization on the symmetry-preserving manifold, not a theorem about every unconstrained optimum of an ordered ansatz; and unitary doubles and classical CCD remain distinct approximations, as the different energies of Table I show.) A variational solver supplies an *approximation* to the coarse ground state,

$|\psi_{\rm low}\rangle = c_0\,|E_0^{\rm low}\rangle + \sum_{j>0} c_j\,|E_j^{\rm low}\rangle$ with coarse infidelity $\epsilon_{\rm low} = 1 - |c_0|^2$; the later stages are written for an exact coarse eigenstate, and Sec. III C states what an imperfect input costs.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the $k_{\rm low}$ space to the *same* occupation pattern in the $k_{\rm high}$ space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighboring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\rm sites})$ gates per dimension [7]. The prolonged state $|\Phi_0\rangle = P\,|\psi_{\rm low}\rangle$ satisfies $\langle\Phi_0|H_{\rm high}|\Phi_0\rangle = \langle\psi_{\rm low}|H_{\rm low}|\psi_{\rm low}\rangle$ exactly *because* the nested bare pairing Hamiltonians used here obey $P^\dagger H_{\rm high} P = H_{\rm low}$ (same level spacing and coupling at both resolutions); the identity does not follow from the definition of a prolongation and can fail for a renormalized or re-parametrized coarse Hamiltonian. For the exact coarse eigenstate the right-hand side is $E_{\rm low}$, for a variational one its variational energy. That the prolonged state also has a *large overlap* with the high-resolution ground state is a separate, empirical property of this pair of Hamiltonians, documented by the overlaps reported in Sec. IV B; preservation of the coarse expectation value does not imply it.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\!\left(\tfrac{\pi s}{2}\right) P(H_{\rm low}-\mu)P^\dagger + \sin^2\!\left(\tfrac{\pi s}{2}\right)(H_{\rm high}-\mu), \qquad s = \frac{t}{T}, \tag{3}$$

evolved by a Suzuki–Trotter product over total time T. The shift μ (here $\mu = E_{\rm low} + 0.6$) places the tracked low-energy states below the large null space of $P^\dagger$ at $s = 0$; provided the tracked branch stays isolated from the rest of the spectrum along the whole path — a condition that the shift alone does not guarantee and that the spectral calculations of Secs. IV B–IV D verify case by case — the level followed from $s = 0$ to $s = 1$ is the high-resolution ground state. The shift is not a global energy offset: writing $\mu PP^\dagger\cos^2(\pi s/2) = \mu\mathbb{1}\cos^2(\pi s/2) - \mu\cos^2(\pi s/2)(\mathbb{1} - PP^\dagger)$ exhibits it as a relative penalty on the complement of the prolonged subspace, which is why changing μ changes the path dynamics while leaving the end-point spectrum untouched. The first term is *not* the coarse Hamiltonian acting on the coarse register alone: since $PP^\dagger$ projects on the states in which the appended single-particle levels are empty,

$$A \equiv P(H_{\rm low}-\mu)P^\dagger = (H_{\rm low}-\mu)\otimes|0\cdots0\rangle\langle0\cdots0|_{\rm app}\,, \tag{4}$$

a coarse-space operator conditioned on the $2(k_{\rm high}-k_{\rm low})$ new qubits being in $|0\rangle$; its circuit and cost are given in Sec. V B. At $s = 0$ the gap above the tracked state is $\min(E_1^{\rm low} - E_0^{\rm low},\, \mu - E_0^{\rm low})$: increasing the shift lowers the embedded coarse states further below the null space, and the gain in initial gap saturates at the coarse excitation gap (for a one-dimensional coarse space, as in the $k = 2$ start used below, it is simply $\mu - E_{\rm low}$). Along the path the instantaneous spectrum is found to stay smooth and gapped in all cases studied here. A gapped path alone does not fix the adiabatic time scale; Ref. [7] shows that when the squared norm of the first-order perturbative correction to the tracked state, $\chi(s) = \sum_{n>0} |\langle n|\partial_s H|0\rangle|^2/(E_n - E_0)^2$, stays uniformly bounded along the path, the adiabatic time scales as $T \sim 1/\Delta E$ rather than as the $O(1/\Delta E_{\rm min}^2)$ of the generic adiabatic theorem [3]. Section IV D evaluates χ for our path family and treats the overlap $p = |\langle\Psi_{\rm high}|\Phi(T)\rangle|^2$ as the figure of merit of the stage. Refinement is unitary and does not purify an imperfect coarse input: if the coarse state carries infidelity $\epsilon_{\rm low}$, ideal adiabatic continuation transports that weight to excited fine states, and the final infidelity of the digitized stage obeys

$$1 - F_{\rm refine} \le \left(\sqrt{\epsilon_{\rm low}} + a_{\rm ad} + \eta\right)^2, \tag{5}$$

where $a_{\rm ad}$ bounds the leakage amplitude of the ideal evolution started from the exact coarse ground state and η the operator-norm error of the product formula (decompose the input into its ground and orthogonal components and apply the triangle inequality to the final component orthogonal to the target). The three sources must share a target ϵ; assigning all of it to the adiabatic term and then adding the digitization error does not certify the same total. The residual excited weight is what the rodeo stage removes, provided the overlap with the target remains nonzero.

The interpolation (3) is the $\sin^2(\pi s/2)$ schedule, whose vanishing derivatives at both ends remove the boundary terms of the adiabatic bound of Jansen, Ruskai and Seiler [11]; it is also the schedule of the nucleation method [9], where each growth stage couples newly added qubits to a bulk lattice in the way the second term of Eq. (3) couples the appended levels to the coarse space.

Ref. [9] turns that bound into explicit sufficient conditions for a Trotterized adiabatic stage: a total time

$$T \;\geq\; \frac{1}{\sqrt{\epsilon}} \int_0^1 \left[\frac{\left\| \ddot{H} \right\|}{\Delta^2} + 7\, \frac{\left\| \dot{H} \right\|^2}{\Delta^3} \right] ds \tag{6}$$

guarantees a final infidelity at most ϵ, with derivatives taken with respect to s and $\Delta(s)$ the instantaneous gap, and $n_s \geq \eta^{-1}(T\Lambda_1/4 + T^2\Lambda_2/2)$ first-order Trotter steps keep the operator error of the digitized evolution below η, with $\Lambda_1 = \max_s \left\| \dot{H} \right\|$ and, for the many groups of our compilation, $\Lambda_2 = \max_s \sum_{i<j} \|[H_i(s), H_j(s)]\|$, the sum of the commutator norms over all pairs of groups — the natural extension of the two-group statement of Ref. [9], which is what Sec. IV E evaluates. The time condition applies verbatim to Eq. (3), since the path is twice differentiable; but both conditions are sufficient, not necessary, and Ref. [9] documents for a twelve-site transverse-field Ising chain that they overestimate the time and the step count actually needed by several orders of magnitude — in part because the Trotter error of a digitized adiabatic sweep partly heals itself [12]. The refinement times T^* extracted below are therefore empirical, read off the computed overlap with the exact fine ground state rather than from a bound, and the product-formula step counts of Sec. V B are chosen by the same exact benchmark. The refinement circuit is optimizer-free in execution — no parameter is trained — but its controls (T, μ, n_s) are calibrated here against the exact answer; an implementable calibration would read them from the acceptance statistics of the filter or from a coarse-space estimate of χ, and we have not demonstrated that. Sec. IV E evaluates both conditions for our path and quantifies the overshoot.

D. The rodeo algorithm

The rodeo algorithm [5, 6] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement; the circuit is shown in Fig. 12): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(E\,t_k)$ on the ancilla (which effectively shifts $H \to H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}\big(\mathbb{1} + e^{iEt_k} e^{-iHt_k}\big)$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2}\cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla measurements return 0 is therefore [6]

$$P_{0^M}\big(E \mid \{t_k\}\big) = \prod_{k=1}^{M} \cos^2\Big[\tfrac{t_k}{2}\big(E_{\rm obj} - E\big)\Big], \tag{7}$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0^M}(E) = \left[\frac{1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}}{2} \right]^M . \tag{8}$$

On resonance ($E = E_{\rm obj}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is $1/2$). Two readings of that background must be kept apart: the Gaussian-averaged single-cycle factor $q = \frac{1}{2}[1 + e^{-(E_{\rm obj}-E)^2\sigma^2/2}]$ is never smaller than $1/2$, so for the *ensemble* of random schedules $q^M \geq 2^{-M}$ is a lower bound on the retained fraction of any component, reached only far from resonance ($|E_{\rm obj} - E|\sigma \gg 1$); for one *fixed* schedule there is no such bound, since an individual cosine factor can vanish. Scanning E locates eigenvalues as peaks whose width is $\sim 1/\sigma$ at fixed M and, since $q(E)^M \simeq \exp\big[-M\sigma^2(E - E_j)^2/4\big]$ near a peak, $\sim 1/(\sigma\sqrt{M})$ when the cycle number is also varied; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realized by a Suzuki–Trotter decomposition over the $2k$ qubits [19–21]; for the single-qubit benchmark of Ref. [6] it reduces to a Z–Y–Z Euler rotation $R_{\hat{n}}(\theta) = e^{-i\theta\,\hat{n}\cdot\vec{\sigma}/2}$ implemented through the Qiskit native gate. In Secs. IV A–IV F we apply the controlled evolution exactly within the conserved N-particle sector to isolate the algorithmic behavior; the circuit-level pipeline with explicit product formulas, and the Trotter error it introduces, are quantified in Sec. V B. Throughout we draw the cycle times t_k from a Gaussian as in Ref. [5]; optimized deterministic [46] or geometric [48] time schedules suppress off-resonant weight faster at fixed runtime and would only improve the numbers reported here, as would the controlled-reversal-gate compilation of the controlled evolution of Ref. [47].

a. Convergence criterion and the random-time ensemble. With E tuned to the target eigenvalue E_0, the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k\,|E_k\rangle$ after M cycles, averaged over the Gaussian time sets, is

$$P_M = p + \sum_{k\neq 0} |c_k|^2\, q_k^M, \qquad p = |c_0|^2, \tag{9}$$

with $q_k = \frac{1}{2}\big(1 + e^{-(E_k-E_0)^2\sigma^2/2}\big)$ the Gaussian-averaged single-cycle factor of Eq. (8). On resonance and with exact evolution the target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \to p$; at a detuning $d = E - E_0$ the retained target weight is $p\,q_0(E)^M$ and decays with M, so a substantial acceptance requires $|d|\sigma\sqrt{M} \ll 1$ (Sec. IV F tests this), and with a product-formula evolution the preservation is only approximate (Sec. V B). For an exactly degenerate target energy the filter selects the eigenspace, not a vector; and a component with zero initial overlap

cannot be created by any function of H. Two fidelities must be distinguished for a randomized filter, whose schedules are drawn afresh for every attempted preparation. For one fixed time set $\{t_k\}$ the post-selected state has a fidelity $\mathcal{F}(\{t_k\})$ and an acceptance probability $P(\{t_k\})$; the fidelity of the *ensemble of accepted runs*, which is what repeated executions on a device deliver, is the acceptance-weighted average $\mathcal{F}_M = \langle P\,\mathcal{F}\rangle/\langle P\rangle$, and for it the spectral decomposition gives exactly

$$1-\mathcal{F}_M = \frac{\sum_{k\neq 0}|c_k|^2 q_k^M}{p+\sum_{k\neq 0}|c_k|^2 q_k^M} \leq \frac{(1-p)\,q_{\max}^M}{p+(1-p)\,q_{\max}^M}, \quad (10)$$

with $q_{\max} = \frac{1}{2}\big(1+e^{-\Delta_{\min}^2\sigma^2/2}\big)$ and $\Delta_{\min}$ the smallest excitation energy carrying weight. The familiar estimate $1-\mathcal{F}_M \simeq \frac{1-p}{p}2^{-M}$ is the limit $\Delta_{\min}\sigma \gg 1$ of Eq. (10); for the pairing spectra used below ($\Delta_{\min}\sigma \geq 5$) the two are indistinguishable, and we plot Eq. (10) as the analytic expectation of the ensemble average. All sampled infidelities in this paper are acceptance-weighted; the unweighted mean of $\mathcal{F}(\{t_k\})$ over time sets differs from $\mathcal{F}_M$ by at most $1/p-1$ relative, i.e. less than one percent for inputs with $p > 0.99$ and a few percent for $p \simeq 0.87$, and the dominant uncertainty of a sampled point is instead the finite number of schedules (Sec. IV F). Reaching an ensemble fidelity $1-\delta$ needs $M \gtrsim \log_{1/q_{\max}}[(1-p)/(p\,\delta)]$ cycles, and since the expected number of repetitions per accepted run is $1/P_M \to 1/p$, the ideal controlled-evolution effort at a resolution ϵ sufficient to isolate the target scales as $O\big(|\log\delta|/(p\,\epsilon)\big)$ [5]. This is not the gate complexity of the implemented pipeline: input preparation, the search for the target energy, the accuracy of the Hamiltonian simulation and the reduction of the product-formula step with the requested precision are counted separately in Sec. V, and the accuracy floor found there is the direct reason for the distinction. The filter is efficient whenever p is close to one, whatever supplies that overlap — resolution refinement, a variational circuit or a cheap correlated reference — which is why the comparison of Sec. V C is between *inputs* to the same filter.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\mathrm{UCCSD}} \geq E_{\mathrm{FCI}}$), tracking FCI to $\sim 10^{-3}$–10^{-2}, whereas CCD over-binds. Figure 1 compares all solvers (exact, CCD, statevector and gate-level UCCSD, and a hardware-efficient ansatz [15]) as a function of k. The hardware-efficient ansatz is exact at $k = 2$ and reaches FCI at $k = 3$ only after a multi-start optimization, and struggles at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit.

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N-sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. "#CNOT" is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	–
8	16	0.145559	0.097263	0.157856	–

Figure 2 shows that the same UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$); the compiled circuits and their depths are analyzed in Sec. V. Two remarks keep this table in proportion. First, an energy error is not an infidelity: for a normalized state $\langle H\rangle - E_0 = \sum_{j>0}|c_j|^2(E_j - E_0) \geq \Delta\,(1-\mathcal{F})$, so with the physical gap $\Delta = 1.823$ the $k = 4$ energy error 1.44×10^{-3} already caps the infidelity at 7.9×10^{-4}; the actual value, obtained from the optimized state, is $1-\mathcal{F}_{\mathrm{UCCSD}} = 1.78\times10^{-4}$, a factor eight below the energy error, and it is this infidelity that every comparison below uses. Second, the scan in k at fixed $N = 4$ is a basis-size study — the fixed-N sector grows as k^4 and the seniority-zero sector as k^2 — and it concerns the UCCSD solver alone, not the complete pipeline; it is not evidence of favorable many-particle scaling at fixed filling.

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917 for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to $\simeq 0.999$ and the energy falls from $E_{\mathrm{HF}} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. 4). Energies above FCI are guaranteed by the variational principle for any normalized state; the improvement with T is what is computed, and monotonicity at every time is not implied. The "exact-slice" propagation still approximates the continuously changing Hamiltonian by 160 midpoint slices; refining to 1 280 slices changes the overlap by 1.5×10^{-6} at $T = 25$ (0.9975713 → 0.9975698) and by 1.1×10^{-6} at $T = 30$, far below the adiabatic infidelities of order 10^{-3} discussed below. The instantaneous spectrum stays open along the whole path (Fig. 3, right), which makes adiabatic reasoning applicable but does not by itself select the $T \sim 1/\Delta E$ cost law dis-

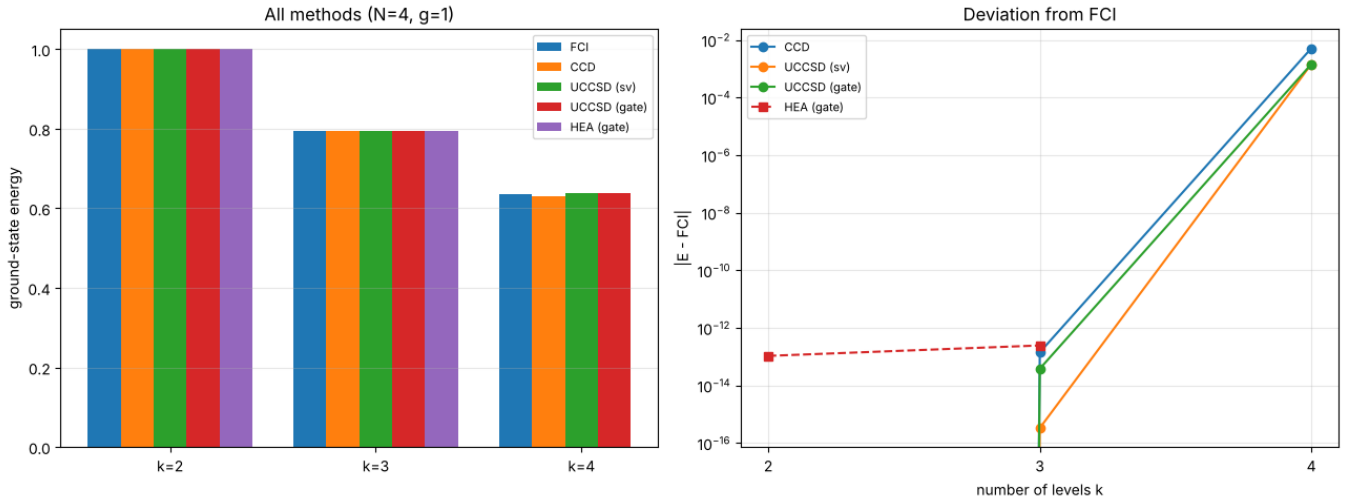


FIG. 1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels k. The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

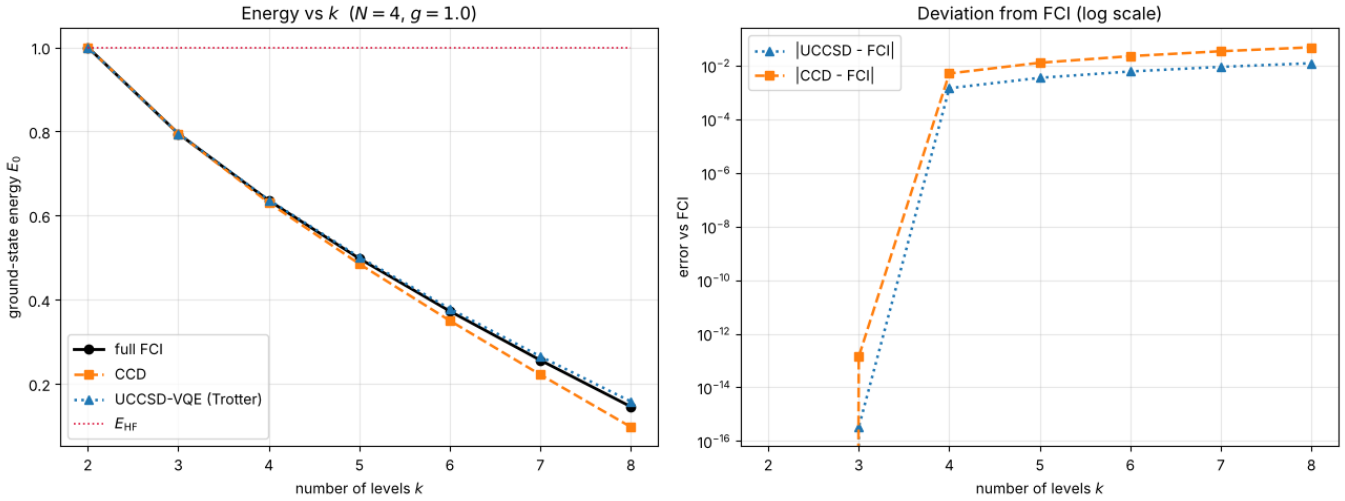


FIG. 2. Scalability of the Trotterized UCCSD–VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

cussed in Sec. III C; the larger $k = 4$ space starts lower in overlap and converges more slowly.

C. A nontrivial coarse stage: the $k = 3 \to 4$ chain

At $N = 4$ the $k = 2$ sector is a single determinant, so in the chain above the coarse "solver" is gate-free and the pipeline's first stage does no work. To exercise the pipeline with a working coarse stage we start instead from the gate-level UCCSD state at $k = 3$ — a genuine 272-CNOT variational preparation, indistinguishable from FCI at this resolution (Table I) — prolong $3 \to 4$, and refine along the same path, with the shift μ now referenced to the *variational* coarse energy $\langle\psi_{\rm low}|H_{\rm low}|\psi_{\rm low}\rangle$, the only quantity a device can know. Figure 5 compares the two chains. The better coarse start raises the prolonged overlap from 0.868 to 0.964 and reaches the target overlap 0.99 at $T^* = 8$ (overlap 0.9907; the grid of Fig. 6 gives 9) instead of $T^* = 16$. The $k = 3$ coarse state is essentially exact ($1 - F < 10^{-9}$), so this chain exercises a working coarse stage but does not test the pipeline against a non-negligible coarse-solver error. Whether the shorter sweep saves gates is decided by the embedded coarse term of Eq. (4), which is not free when the coarse space is non-trivial. For the $2 \to 4$ chain A is a single four-qubit conditional phase (14 CNOTs per application, Sec. V B); for the $3 \to 4$ chain it is the complete $k = 3$ Hamiltonian conditioned on the two appended qubits being empty, about 350 CNOTs per application, more than the 296

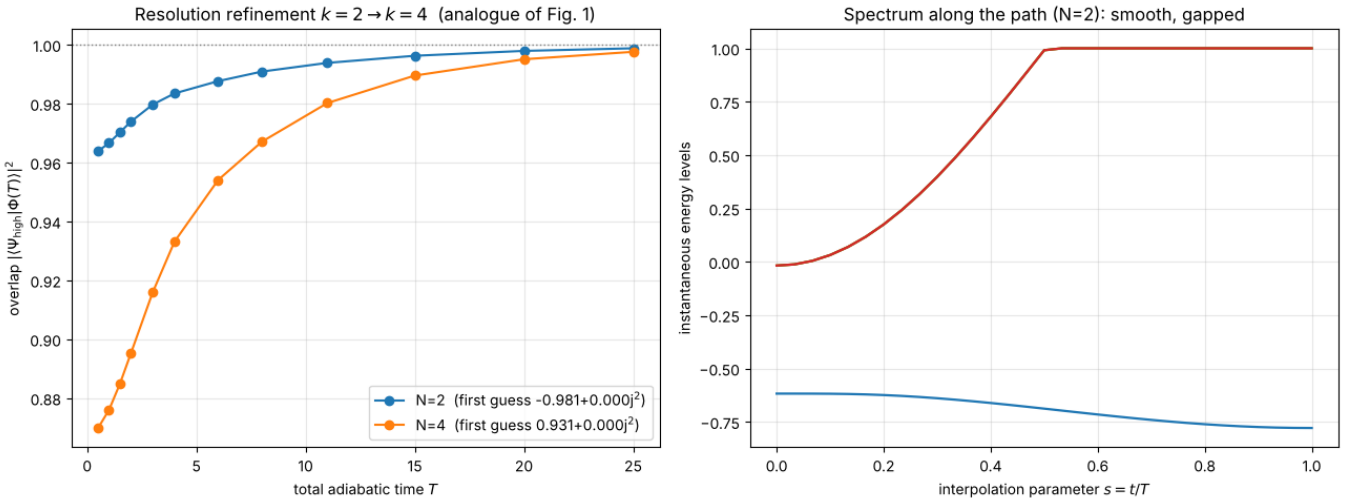


FIG. 3. Resolution refinement $k = 2 \to k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T, for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

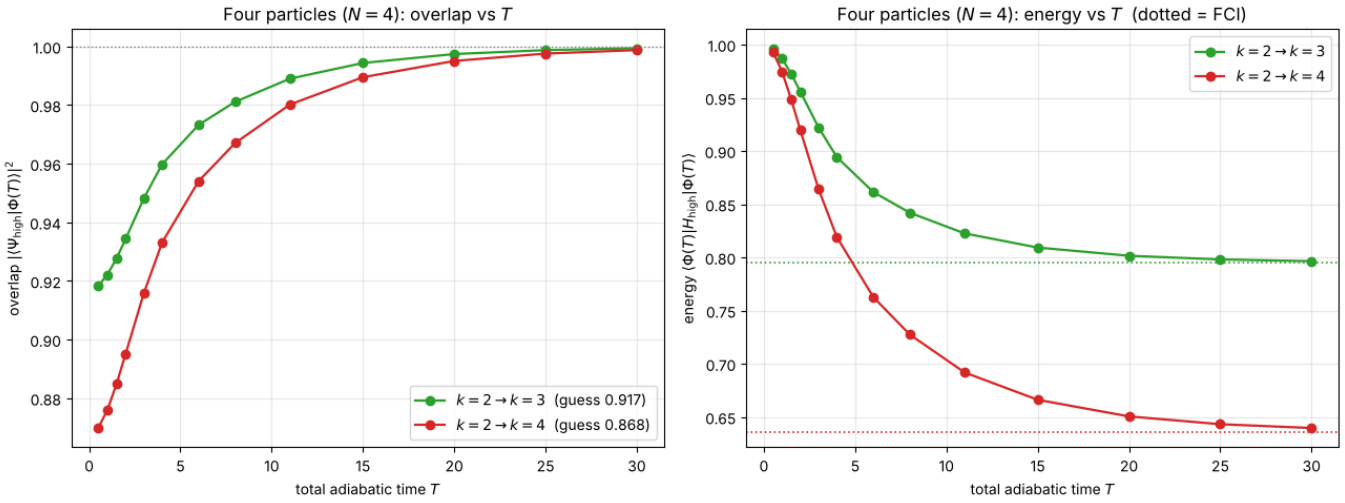


FIG. 4. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle\Phi(T)|H_{\text{high}}|\Phi(T)\rangle$ (right) versus T for $k_{\text{high}} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity, both on a $\sim 1/\Delta E$ timescale.

CNOTs of the high-resolution pairing step itself. Because H_{low} contains non-commuting terms, the counted circuit does not exponentiate A as one block but splits it into its own coarse groups (the conditioned diagonal term and three conditioned pair-hopping terms), and we validate the chain at that level in Sec. V B: 22 second-order steps at $T = 8$ give $p = 0.9912$, so the $3 \to 4$ chain costs $2 \times 22 \times (296 + 350) + 96 \simeq 2.9 \times 10^4$ CNOTs, the 96 being the coarse preparation (two pair rotations prepare the three-dimensional seniority-zero coarse space exactly; the 272-CNOT UCCSD template prepares the same state). At the step density of Table III the $2 \to 4$ chain needs 43 steps for $T^* = 16$, i.e. $2 \times 43 \times (296 + 14) \simeq 2.7 \times 10^4$ CNOTs. The shorter adiabatic time of the better start is thus offset by the cost of carrying the coarse Hamiltonian along the path, and the two chains cost the same within 10%; neither figure is an optimized minimum. Starting higher in the hierarchy remains the right choice when the prolonged overlap of the trivial start is poor (the strong-coupling regime of Sec. IV D), but it is not a gate saving in itself.

D. Coupling scan and the path gap

All results above sit at $g = 1$; Fig. 6 probes the refinement across couplings and tests the $T \sim 1/\Delta E$ claim directly. Scanning $g = 0.3$–4 at fixed $\mu = E_{\text{low}} + 0.6$

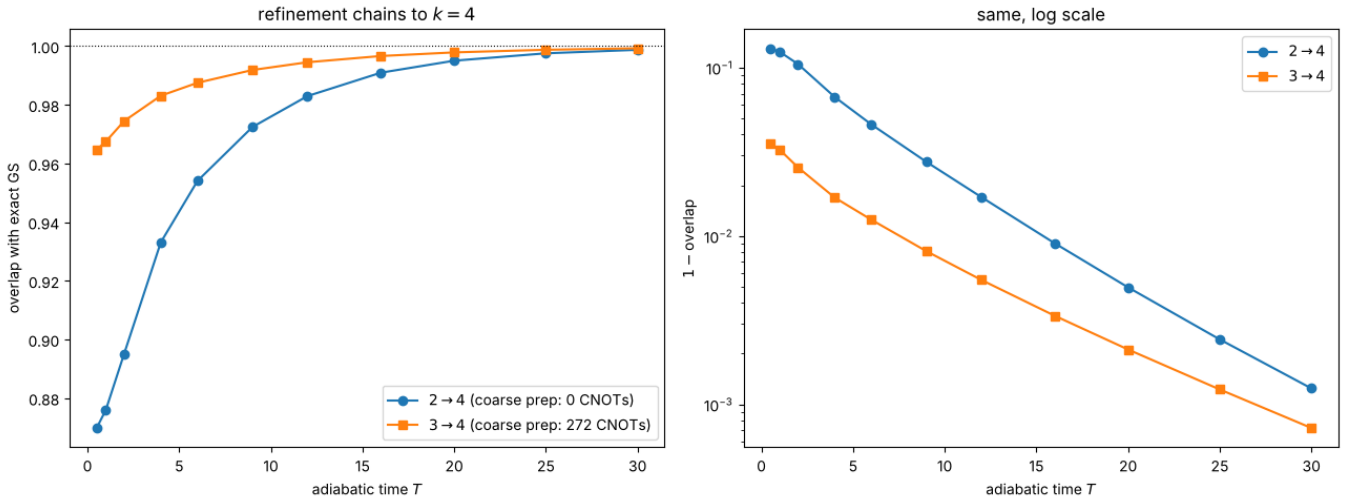


FIG. 5. Refinement chains to $k = 4$ at $N = 4$, $g = 1$, starting from the gate-free $k = 2$ determinant (circles) and from the 272-CNOT gate-level UCCSD state at $k = 3$ (squares). Left: overlap with the exact ground state versus adiabatic time T; right: the same on a logarithmic scale. The $3 \to 4$ chain starts at overlap 0.964 (versus 0.868) and reaches 0.99 in half the adiabatic time ($T^* = 8$ on a finer grid than the one shown).

(left panel), the minimum instantaneous gap along the path stays open and nearly constant — it is set by the shift itself, $\Delta E_{\min} \simeq \mu - E_{\text{low}}$, dropping only to 0.57 at $g = 4$ — while the time to reach overlap 0.99 grows from $T^* = 1$ at $g = 0.3$ to $T^* = 65$ at $g = 4$. The growth goes together with the loss of initial overlap of the prolonged state, which falls from 0.99 to 0.38 over the same range, consistent with a larger change of the eigenstate along the path; but changing g also changes the transition matrix elements and the eigenvector path, so the scan does not isolate the overlap as the cause (the shift scan below changes T^* at fixed initial overlap). It does establish that the nearly constant minimum gap does not determine the required time by itself, which identifies the practical limiter at strong coupling as the starting state rather than a closing gap; whether a coarse start higher in the hierarchy (Sec. IV C) is then the better route is a question of total cost, including the coarse preparation and its conditioned Hamiltonian, not of sweep time alone.

Because the path gap is controlled by the shift, varying μ at fixed $g = 1$ changes the minimum gap over a factor five (right panel). The four finite threshold times, $(\Delta E_{\min}, T^*) = (0.3, 50), (0.6, 16), (1.0, 8), (1.5, 6)$, are fitted with an intercept by $T^* \approx 17.1/\Delta E_{\min} - 8.4$ (sum of squared residuals 28) and, equally legitimately, by $T^* \approx 4.1/\Delta E_{\min}^2 + 4.2$ (residuals 0.2); four threshold times read off a coarse time grid, along a family of paths whose operator changes with μ, do not identify an exponent, and in particular do not exclude inverse-square behavior. We therefore present the scan as a sensitivity study. The response norm χ of Sec. III C makes the limitation explicit (Fig. 7, right). With $u = \sin^2(\pi s/2)$ the path is $H(u) = (1-u)A + uB$ with $A = -b\,|\text{HF}\rangle\langle\text{HF}|$, $b = \mu - E_{\text{low}}$, and at $u = 0$ the HF determinant couples through $\partial_u H$ to four pair-transfer configurations with amplitude $-g/2$, all at excitation energy b, so $\chi_u(0) = g^2/b^2$ exactly: 11.1 at $b = 0.3$ and 0.44 at $b = 1.5$, a factor 25, and the maximum of $\chi_u(s)$ sits at $s = 0$ for every shift. The quantity that the inverse-gap argument of Ref. [7] assumes uniformly bounded is thus itself varied by the shift scan, which cannot then test that argument by attributing the timing changes to the gap alone. Two further gaps should be kept apart from the artificial shift-controlled gap at $s = 0$: the physical gap of H_{high} in the $N = 4$ sector (1.823 at $g = 1$), a conservative all-sector bound, and the dynamically accessible seniority-zero excitation gap at $s = 1$ (2.300), which is the one the transition matrix elements see. The same scan doubles as the sensitivity study for the shift: the choice $\mu_{\text{buf}} = 0.6$ used throughout is conservative, and enlarging the buffer to 1.5 *shortens* the refinement to $T^* = 6$. For a one-dimensional coarse space there is no upper limit to the buffer other than the growth of the energy scale (and hence of the Trotter error) of the path; for coarse spaces with several states the useful initial gap saturates at the coarse excitation gap, as noted in Sec. III C. Section V C adds the limiting case of this observation: an ordinary adiabatic ramp from the diagonal (HF) Hamiltonian, which has no shift and whose seniority-zero gap at $s = 0$ is 2.0, refines the same determinant in a quarter of the time.

E. Sufficient conditions versus practice

The conditions quoted in Sec. III C can be evaluated exactly for the path of Eq. (3). Writing $A = P(H_{\text{low}} - \mu)P^\dagger$ and $B = H_{\text{high}} - \mu$, the schedule gives $\dot{H} = \frac{\pi}{2}\sin(\pi s)\,(B - A)$ and $\ddot{H} = \frac{\pi^2}{2}\cos(\pi s)\,(B - A)$,

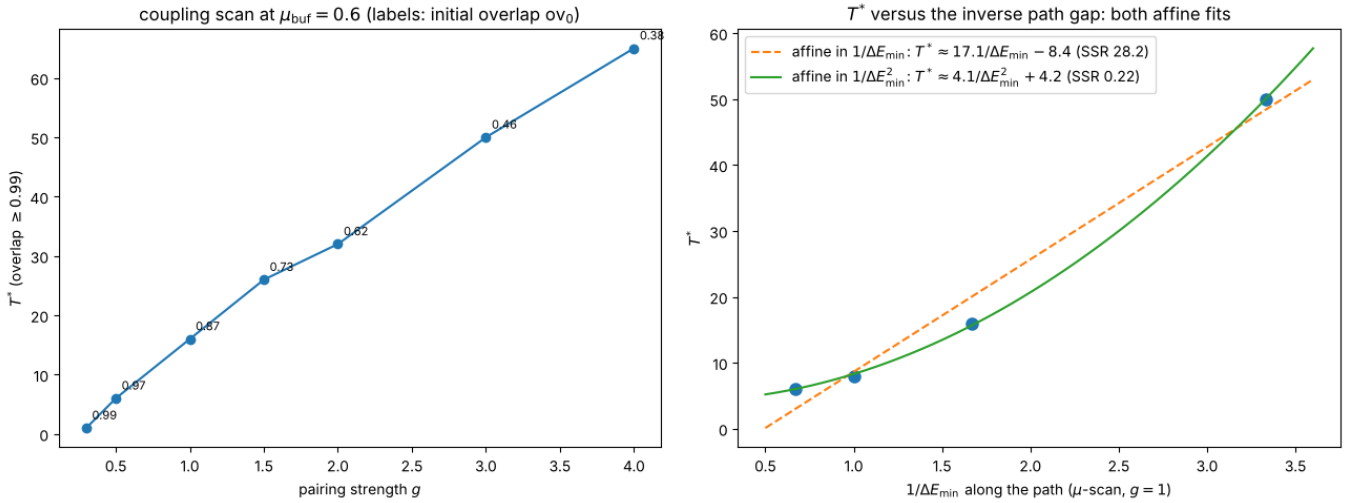


FIG. 6. Left: adiabatic time T^* to reach overlap 0.99 for the $k = 2 \to 4$ refinement versus pairing strength g (point labels: initial overlap of the prolonged state). The minimum path gap stays $\simeq 0.6$ throughout while the starting overlap degrades. Right: T^* versus the inverse minimum path gap, varied through the shift buffer $\mu_{\rm buf} = 0.3$–1.5 at $g = 1$, with affine fits in $1/\Delta E_{\rm min}$ (dashed) and in $1/\Delta E^2_{\rm min}$ (solid); four points on a coarse time grid fit either form and identify no exponent.

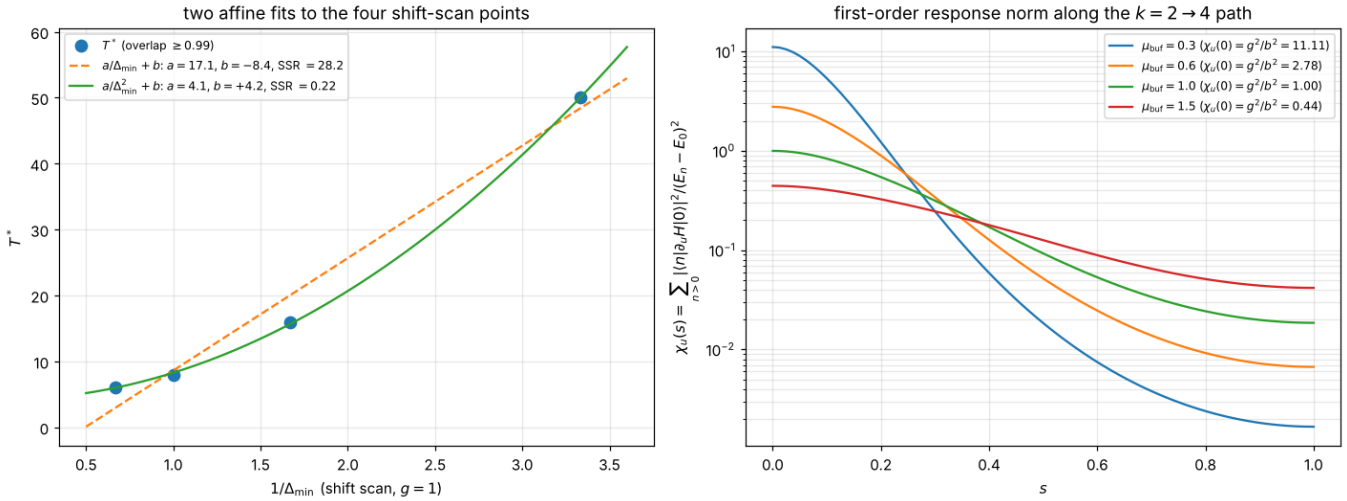


FIG. 7. Left: the four shift-scan points of Fig. 6 with the two affine fits, in $1/\Delta E_{\rm min}$ (dashed) and in $1/\Delta E^2_{\rm min}$ (solid), and their sums of squared residuals. Right: the first-order response norm $\chi_u(s) = \sum_{n>0} |\langle n|\partial_u H|0\rangle|^2/(E_n - E_0)^2$ of the $k = 2 \to 4$ path, $u = \sin^2(\pi s/2)$, for the four shifts; at $s = 0$ it equals g^2/b^2 with $b = \mu - E_{\rm low}$ and it is largest there, so the quantity that the inverse-gap argument assumes bounded changes by a factor 25 across the scan.

so the integrand of Eq. (6) is fixed by the single number $\|B - A\|$ and by the instantaneous gap $\Delta(s)$, both taken in the $N = 4$ sector in which the evolution takes place. Figure 8 shows the two along the $k = 2 \to 4$ path: the gap rises from the shift-controlled 0.600 at $s = 0$ to the physical gap of $H_{\rm high}$ at $s = 1$, and the integrand peaks near $s \simeq 0.25$, where $\sin(\pi s)$ is already sizable while the gap is still close to its minimum; the term $7\left\|\dot{H}\right\|^2/\Delta^3$ supplies 94% of the integral. Table II collects the numbers. Here $\|B - A\| = 7.62$ and $\int_0^1 [\cdots]\, ds = 511$, so Eq. (6) demands $T \geq 5.1 \times 10^3$ for an infidelity of 10^{-2} (and 1.6×10^4 for 10^{-3}), whereas the computed overlap reaches 0.99 at $T^* = 16$: the sufficient condition overshoots the practical time by a factor 3×10^2. The step-count condition is looser still. With $\Lambda_1 = \max_s \left\|\dot{H}\right\| = 12.0$ and, for the term partition of the gate compilation of Sec. V B (the conditioned coarse term, the diagonal term and the six pair-hopping terms), $\Lambda_2 = \max_s \sum_{i<j} \|[H_i(s), H_j(s)]\| = 13.0$, the first-order bound at $\eta = \sqrt{\epsilon}/2$ requires 3.4×10^4 steps along the $T^* = 16$ path — and 3.4×10^9 along the path of length $T_{\rm b}$ — while 160 first-order or 40 second-order steps al-

ready reproduce the exact-slice overlap at T^* to within the adiabatic error of the path itself. The two step counts answer different questions and the ratio between them must be read with that in mind: the bound certifies an operator-norm error of the digitized unitary, whereas reproducing one scalar overlap does not certify that the two output vectors, let alone the two unitaries, are close. (The companion paper [10] repeats this evaluation with the particle-hole term switched on, where the time bound is insensitive to the pair breaking because the minimum gap is set by the shift, while the step bound grows with the number of Trotter groups.)

These ratios — 3×10^2 for the time and 2×10^2 for the step count — are the fermionic counterpart of what Ref. [9] reports for nucleation of a twelve-site Ising chain (6×10^1 and 10^7, respectively), and they carry the same message: the conditions certify convergence but do not predict cost. Part of the reason is visible in the integrand. The bound pays for the full operator norm $\|B - A\| = 7.62$ — the same value in the full $N = 4$ sector and in the six-dimensional seniority-zero block that the dynamics actually explores, so it is not an artefact of inaccessible sectors — and for the global gap, whereas the adiabatic error is governed by the individual matrix elements of dH/ds between the tracked state and the states it couples to, relative to their individual gaps. Those elements are small but not zero for the highest accessible states: at $s = 0.25$ the derivative couplings from the ground state to the five seniority-zero excited states are 0.60, 0.37, 0.00, 0.11 and 0.08. A global norm therefore overestimates the cost by ignoring the state-specific structure of the couplings; the magnitude of the overshoot alone does not single out one cause, nor does it verify the inverse-gap mechanism of Ref. [7].

F. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathbb{1} + e^{iEt}e^{-iHt})$ to 10^{-16}, and the all-zero probabilities reproduce Eqs. (7)–(8), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E\,\sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B at $T = 25$ with exact slices and exact controlled evolution, with overlap $p = 0.9976$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). This is an *ideal* operating point, chosen to exhibit the filter; the digital pipeline of Sec. V uses other refinement times and a product-formula evolution, and Table IV keeps the two apart. Scanning the target energy E (Fig. 9; $M = 4$ cycles, 12 time sets per energy) produces a peak at the ground-state energy; a coarse scan ($\sigma = 2$, 60 energies on $[-0.5, 4]$) resolves it from the excited band, and a fine scan ($\sigma = 8$, 161 energies with spacing 0.0075 on a window of width 1.2 centred on the highest point of the coarse scan, $E = 0.644$, i.e. chosen without knowledge of E_0) locates it at $E = 0.6362$ against the exact $E_0 = 0.635548$ by three-point parabolic interpolation at the grid maximum. On a device every acceptance is a shot estimate: adding binomial noise for 10^3 and 10^4 shots per energy and locating the peak by a quadratic fit through the points above 80% of the maximum gives 0.637 ± 0.001 and 0.6373 ± 0.0004, so with this scan the systematic error of the peak estimator ($\sim 2 \times 10^{-3}$, from the non-parabolic shape of the peak over the fitted window) exceeds the shot noise, and the agreement to 10^{-3} is a property of the scan design rather than a demonstrated four-decimal precision. Starting from the refined state, essentially all spectral weight sits in the single ground-state peak; starting from Φ_0, the small residual weight on excited states is visible as secondary structure above the 2^{-M} background. The scan is part of the cost of the method: the fine scan alone evaluates 161 energies with 12 time sets each, i.e. about 2×10^3 four-cycle runs, each repeated over enough shots to estimate an acceptance probability.

Fixing $E = E_0$ and applying M cycles with $\sigma = 3$ (Fig. 10; 40 Gaussian time sets, the same sets for both inputs, generator seed 7 reset per input), the fidelity rises geometrically and the cumulative acceptance settles at $\approx p$, in agreement with Eqs. (9)–(10). For the refined input the exact Gaussian-ensemble infidelity of Eq. (10) is 1.2×10^{-3} after one cycle, 6.1×10^{-4} after two, 3.8×10^{-5} after six and 2.4×10^{-6} after ten, and 12 cycles are needed to reach 10^{-6}; the 40-schedule sample gives 1.35×10^{-3}, 6.5×10^{-4}, 2.1×10^{-5} and 2.0×10^{-6}. The sample lies 45% below the ensemble value at six cycles, which is ordinary finite sampling of products of random attenuation factors: the relative standard error of a 40-schedule mean of the residual weight, computed from 2×10^4 schedules, is 13% at $M = 2$, 37% at $M = 6$ and 82% at $M = 10$ (the error bars of Fig. 10), so the sampled points are statistically consistent with the curve but carry no precision at the smallest infidelities, and the quoted infidelities of this paper are the ensemble values unless stated otherwise. (The acceptance weighting itself changes the sampled infidelity by at most $1/p - 1 = 0.24\%$ at this p; an earlier version of the notebook advanced one generator through both inputs and reported 5.5×10^{-6} at ten cycles, an equally valid sample.) The prolonged input reaches 1.3×10^{-4} after ten cycles (ensemble value 1.5×10^{-4}) while its acceptance holds at 0.868. Each cycle buys roughly a factor two in infidelity, so reaching 10^{-6} from $p = 0.998$ takes of order ten cycles, not one or two. Two practical checks complete the picture. First, the target energy on a device is the scan estimate, not the exact eigenvalue: repeating the filter at the scan estimate ($E - E_0 = 7 \times 10^{-4}$, $|d|\sigma\sqrt{M} = 0.005$ at six cycles) changes no digit of the ensemble infidelities quoted above. Second, a deliberately detuned target attenuates the ground state as well: for $E - E_0 = 0.05$, 0.1 and 0.2 the six-cycle ensemble infidelity is unchanged within 3%, 14% and a factor 1.7, while the acceptance falls from

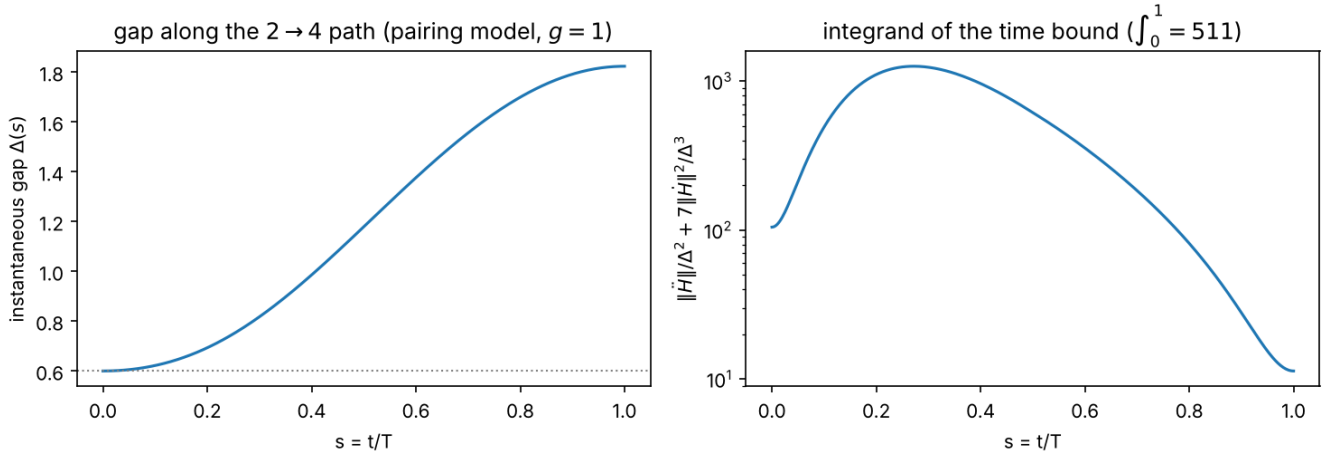


FIG. 8. The ingredients of the adiabatic-time bound, Eq. (6), along the $k = 2 \to 4$ refinement path at $N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$. Left: instantaneous gap $\Delta(s)$ in the $N = 4$ sector, from the shift-controlled 0.600 at $s = 0$ (dotted line) to the physical gap 1.823 of $H_{\rm high}$ at $s = 1$. Right: the integrand $\left\|\ddot{H}\right\|/\Delta^2 + 7\left\|\dot{H}\right\|^2/\Delta^3$ (logarithmic scale), which peaks near $s \simeq 0.25$ and integrates to $I = 511$ (Table II).

TABLE II. Sufficient conditions of Ref. [9] evaluated for the $k = 2 \to 4$ refinement path ($N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$; spectral norms in the $N = 4$ sector) against the empirical cost. $\Delta_{\min}$: minimum instantaneous gap; I: the integral in Eq. (6); $T_{\rm b}$: the time bound for infidelity $\epsilon = 10^{-2}$, $T_{\rm b} = I/\sqrt{\epsilon}$; T^*: time for overlap 0.99 (grid of Fig. 6); $\Lambda_1 = \max_s \left\|\dot{H}\right\|$ and $\Lambda_2 = \max_s \sum_{i<j} \|[H_i, H_j]\|$ over the Trotter groups of the gate compilation; $n_{\rm b}(T^*)$: first-order Trotter steps required by the bound along the T^* path at $\eta = \sqrt{\epsilon}/2$; n_1, n_2: first- and second-order steps (from the sequence $5, 10, 20, \ldots$) that reproduce the exact-slice overlap at T^* to within the adiabatic error $1 - p(T^*)$ of the path.

$\Delta_{\min}$	$\|B - A\|$	I	$T_{\rm b}$	T^*	$T_{\rm b}/T^*$	Λ_1	Λ_2	$n_{\rm b}(T^*)$	n_1	n_2
0.600	7.62	511	5.1×10^3	16	3.2×10^2	12.0	13.0	3.4×10^4	160	40

0.998 to 0.965, 0.873 and 0.596 (exact ensemble values, Sec. 19 of the notebook; an earlier version quoted sampled values) — in this regime the acceptance, not the fidelity, is the sensitive diagnostic of a wrong target, and it then no longer tends to the input overlap; a detuning towards another eigenvalue would instead select that state, which the acceptance alone does not certify against.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

All quantum operations in this work are compiled to a common gate set of single-qubit rotations and CNOTs, so the resource costs of the different routes to the high-resolution ground state — a full-space UCCSD–VQE, the refinement-plus-filter pipeline, and the filter applied directly to cheaper inputs — can be compared gate by gate. Every number quoted below is produced in the accompanying notebook. Two levels of simulation must be distinguished. The Pauli-rotation primitive, the UCCSD excitation circuits, the hardware-efficient ansatz and the ancilla-based rodeo cycle are simulated *gate by gate* on the full $2k$-qubit register and verified against exact exponentials. The product-formula evolutions of Sec. V B, by contrast, are simulated *block by block*: each term of the Hamiltonian splitting is a set of mutually commuting Pauli strings whose exponential is realized exactly by the corresponding rotation–CNOT block, and we apply that exact block exponential within the conserved N-particle sector. This reproduces faithfully the state that the compiled circuit would produce, and the CNOT counts are those of the compiled blocks, but it does not execute every individual rotation and CNOT; we therefore quote CNOT *counts* rather than scheduled circuit depths, which depend on the connectivity and parallelism of a device.

A. Circuit building blocks

The elementary primitive is the Pauli rotation $e^{-i\frac{\phi}{2}P}$ for a Pauli string $P = \bigotimes_q \sigma_q$ [34]: each active qubit is rotated into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT ladder accumulates the parity onto the last active qubit, a single $R_z(\phi)$ is applied, and the ladder and basis changes are uncomputed. A string of weight w therefore

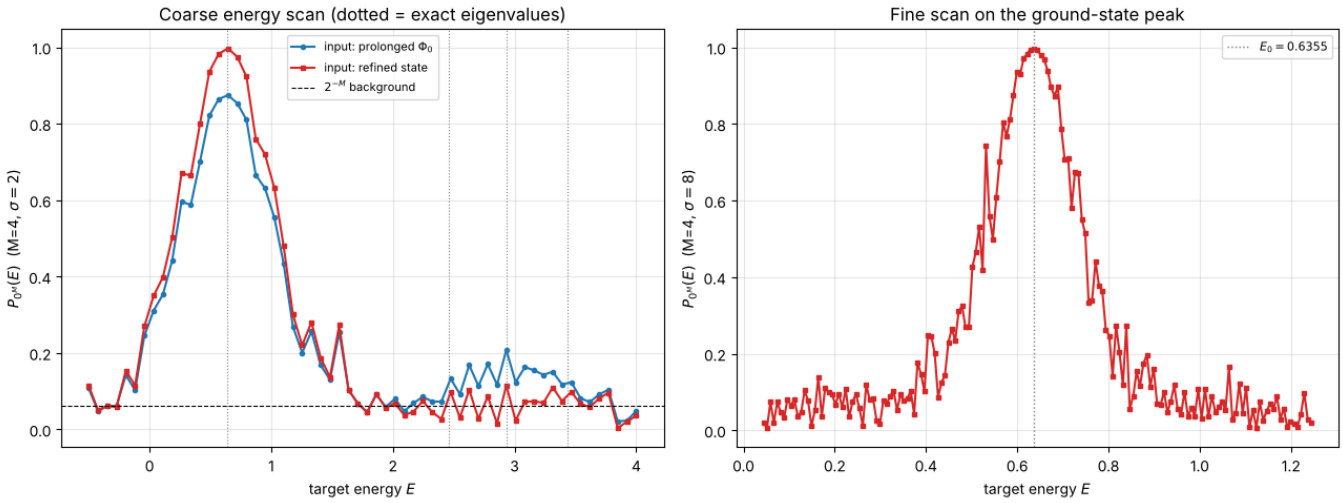


FIG. 9. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on a window centred on the coarse-scan maximum; the peak is located at $E = 0.6362$ (exact $E_0 = 0.6355$).

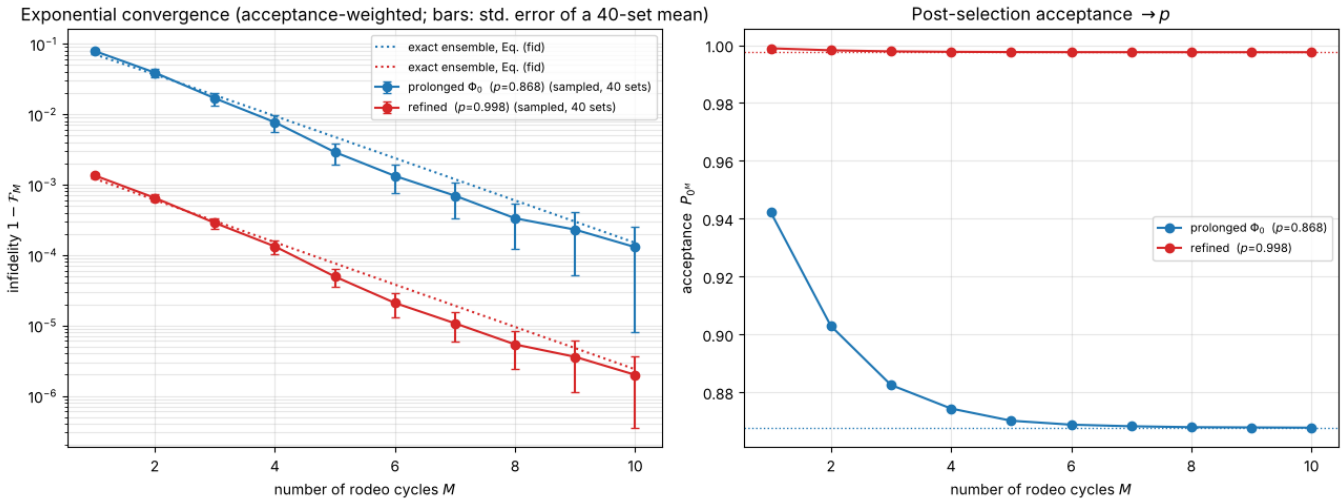


FIG. 10. Rodeo eigenstate preparation at $E = E_0$ ($k = 4$, $N = 4$, $g = 1$, $\sigma = 3$), averaged over 40 Gaussian time sets, for the resolution-refined input ($T = 25$, exact slices, $p = 0.9976$) and the prolonged determinant ($p = 0.868$), with exact controlled evolution and the same 40 time sets for both inputs. Left: acceptance-weighted infidelity of the accepted runs versus the number of cycles M, with error bars giving the standard error of a 40-schedule mean (from 2×10^4 schedules); dotted: the exact Gaussian-ensemble expectation of Eq. (10). Right: the post-selection acceptance converges to the input overlap p.

costs $2(w-1)$ CNOTs and one parametrized rotation; the circuit reproduces the exact exponential to 3×10^{-16}.

a. UCCSD preparation circuit. Through the Jordan–Wigner map a single excitation compiles into 2 Pauli rotations whose weight grows with the distance between the spin orbitals (8 CNOTs for adjacent levels, 16 for the $0 \to 4$ single at $k = 3$), and a generic double excitation into 8 rotations (48–64 CNOTs). The doubles that dominate the pairing problem are special: for a time-reversed pair moving between levels p and q the operators act on the adjacent qubits $(2p, 2p{+}1)$ and $(2q, 2q{+}1)$, so the internal Jordan–Wigner Z strings cancel and all 8 Pauli strings have weight 4 — 48 CNOTs per pair-hopping term *independent of the level separation*. The full state preparation at $N = 4$, $g = 1$ then costs 8 excitation operators, 40 rotations and 272 CNOTs at $k = 3$, and 26 operators, 160 rotations and 1312 CNOTs at $k = 4$ (Table I); at $k = 2$ the ansatz is empty (the sector is a single determinant) and the preparation is gate-free.

Figure 11 shows both levels of this hierarchy. Panel (a) is the VQE at the ansatz level: the product of excitation unitaries acts on the HF register, the energy is estimated from repeated measurements, and a classical

optimizer updates the P amplitudes $\boldsymbol{\theta}$ before the circuit is run again — it is this feedback loop, wrapped around the full-depth state preparation, that multiplies the 1312-CNOT cost into the total gate budget of the traditional route. Panel (b) is the compiled interior of a single weight-4 Pauli rotation, the sub-block that occurs eight times in every pair-hopping double and 160 times in the $k = 4$ preparation; its only parametrized element is the central $R_z(\phi)$, with ϕ fixed by the amplitude of the parent excitation operator.

b. Hardware-efficient ansatz. The HEA [15] used for contrast consists of L layers of R_y rotations on all $2k$ qubits interleaved with a CNOT entangler (nearest-neighbor pairs plus a wrap-around bond), i.e. $2k(L{+}1)$ variational angles and only $O(2kL)$ CNOTs: 12 angles at $k = 2$ ($L = 2$) and 30 at $k = 3$ ($L = 4$). Its depth is far smaller than UCCSD, but the circuit carries no particle-number or seniority structure; a number-penalty term and multi-start optimization are needed at $k = 3$, and at $k = 4$ the optimizer fails to reach the ground state (Fig. 1). Depth is not the only currency: trainability of the symmetry-blind landscape becomes the bottleneck first.

c. Rodeo cycle. One rodeo cycle (Fig. 12) adds a Hadamard on the ancilla, the controlled evolution e^{-iHt_k}, the ancilla phase $P(Et_k)$ — the element that effectively shifts $H \to H - E$ — a second Hadamard, and a mid-circuit measurement; the single ancilla is reset and reused across the M cycles. The contrast with Fig. 11(a) is the structural point of this section: the rodeo circuit contains *no* variational parameters and closes no optimization loop. The times t_k are drawn once from a Gaussian, the target energy E is a classical setting, and the algorithm is driven entirely by the measurement record — a run is accepted when all M ancilla outcomes read 0, which applies the filter of Sec. III D cycle by cycle. Where the VQE re-executes its full-depth circuit to steer $\boldsymbol{\theta}$, the rodeo circuit is executed as printed, and its only system-size-dependent element is the controlled evolution. The controlled evolution is compiled from the same Pauli-rotation primitive: in each rotation sub-circuit only the central R_z needs to be controlled (if the rotation is absent the basis changes and CNOT ladder cancel), and a controlled R_z costs 2 extra CNOTs. For the pairing Hamiltonian (1) a first-order Trotter step contains the $k(k{-}1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each) and $2k$ single-qubit phases, i.e.

$$N_{\mathrm{CNOT}}^{\mathrm{step}}(k) = 24\,k(k-1) + 2k \tag{11}$$

CNOTs uncontrolled — 296 at $k = 4$ — and $\approx 4.2 \times 10^2$ CNOTs per step once the controls are added. A rodeo cycle with n_T Trotter steps therefore costs $\approx 4.2 \times 10^2\, n_T$ CNOTs at $k = 4$.

B. The Trotterized pipeline

The results of Sec. IV F used exact sector exponentials; on hardware both the refinement slices and the controlled evolution are product formulas, and their error must be quantified at the circuit level. We split the Hamiltonian exactly as the gate compilation does: a diagonal term D (kinetic energy plus the $p{=}q$ pairing term; single-qubit phases and ZZ rotations) and one pair-hopping term h_{pq} per level pair. The eight Pauli strings of each h_{pq} commute, so every factor $e^{-ih_{pq}\delta t}$ is realized *exactly* by its 48-CNOT block, and the only Trotter error is the non-commutativity between terms — exactly as on a device. At $k = 4$, $t = 1$ the second-order (symmetric) formula reaches spectral-norm error 3.4×10^{-2} with 4 steps and 8.3×10^{-3} with 8.

a. The embedded coarse term. The refinement path, Eq. (3), contains a third kind of factor, $e^{-i\cos^2(\pi s/2)A\,\delta t}$ with $A = P(H_{\mathrm{low}} - \mu)P^\dagger$ of Eq. (4). Gate-free prolongation does not make this evolution gate-free. For the $2 \to 4$ chain at $N = 4$ the coarse space is the single determinant $|\mathrm{HF}\rangle$, so $A = (E_{\mathrm{low}} - \mu)\,|\mathrm{HF}\rangle\langle\mathrm{HF}| = -0.6\,|\mathrm{HF}\rangle\langle\mathrm{HF}|$, and within the $N = 4$ sector $|\mathrm{HF}\rangle$ is the unique state with the four appended qubits 4–7 all in $|0\rangle$. Its exponential is therefore a phase conditioned on four qubits being empty, $X^{\otimes 4}\,C^3P(\phi)\,X^{\otimes 4}$ with $\phi = 0.6\cos^2(\pi s/2)\delta t$. Expanding the projector, $|1111\rangle\langle 1111| = 2^{-4}\sum_S (-1)^{|S|} Z_S$ over the 15 nonempty subsets S of the four qubits, gives a phase polynomial of 15 commuting Z-strings; ordered along a Gray code it compiles into 14 CNOTs and 15 R_z rotations (the naive ladder-per-string construction costs 34 CNOTs). We verified that this operator reproduces the sector matrix exponential of A exactly. For a chain that starts from a nontrivial coarse space the term is $(H_{\mathrm{low}} - \mu)$ on the coarse register conditioned on the appended qubits being empty; for the $3 \to 4$ chain each of the 33 Pauli rotations of a $k = 3$ step acquires a double control (6 CNOTs per rotation with a three-qubit Gray-code phase polynomial), for about 350 CNOTs per application, which is the figure used in Sec. IV C. All refinement budgets below include the coarse term; for the $2 \to 4$ chain it adds 14 CNOTs to the 296 of a first-order step, i.e. $2 \times 80 \times 14 = 2\,240$ CNOTs or 5% over the $T = 30$ path.

Figure 13 shows the two pipeline stages under this compilation. For the refinement (left), $n_s = 80$ second-order steps along the $T = 30$ path reproduce the exact-slice result to within its own adiabatic error, delivering $p = 0.9977$ against the exact-slice 0.9988; first-order steps need several hundred. For the rodeo stage (right) the honest scheme fixes the maximum Trotter step *size* δt, so a cycle with Gaussian time t_k uses $n_k = \lceil |t_k|/\delta t \rceil$ steps of size t_k/n_k ($\sigma\sqrt{2/\pi}/\delta t \approx 13$ steps on average at $\sigma = 4$, $\delta t = 0.25$). The filter is strikingly diagnostic of its own Trotter error: at $\delta t = 1$ the corrupted evolution is largely *rejected* — the acceptance collapses to 0.10 after six cycles and the fidelity does not approach the target accuracy in the displayed range — while at $\delta t = 0.25$

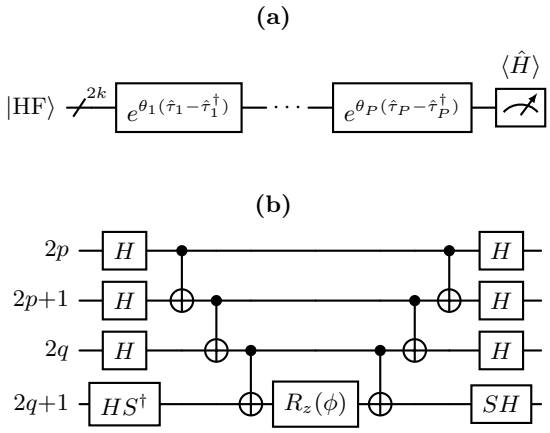


FIG. 11. The variational quantum eigensolver as a circuit. (a) Ansatz level: the Trotterized UCCSD product acts on the HF register ($2k$ qubits, slashed wire) and the energy is estimated by measuring $\langle \hat{H} \rangle$ (in practice a sum of Pauli expectation values, each averaged over many shots); a classical optimizer updates $\boldsymbol{\theta}$ and the entire circuit is re-executed, once per energy or gradient evaluation. (b) Compiled circuit of one weight-4 Pauli rotation, here $e^{-i\frac{\phi}{2}X\otimes X\otimes X\otimes Y}$ acting on the four qubits of a pair-hopping double excitation $(2p, 2p{+}1) \to (2q, 2q{+}1)$: basis changes into the Z eigenbasis (H for X, $HS^\dagger$ for Y), a CNOT parity ladder, the single parametrized $R_z(\phi)$, and the uncompute. Eight such blocks (48 CNOTs) build one double excitation; 160 of them (1312 CNOTs) build the $k = 4$ preparation of Table I.

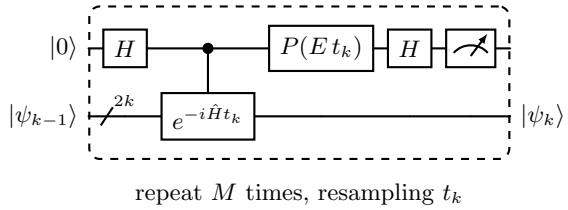


FIG. 12. One cycle of the rodeo algorithm: Hadamard on the ancilla, controlled time evolution $e^{-i\hat{H}t_k}$ of the $2k$-qubit system register with a Gaussian-random time t_k, ancilla phase $P(Et_k)$, second Hadamard, and a mid-circuit measurement of the ancilla, which is then reset and reused for the next cycle. A run is accepted when all M outcomes read 0; each accepted cycle applies the filter $\frac{1}{2}(\mathbb{1} + e^{iEt_k}e^{-i\hat{H}t_k})$ to the system. Unlike the VQE of Fig. 11(a), the circuit carries no variational parameters and no classical optimization loop; the controlled evolution, compiled from the same Pauli-rotation blocks as Fig. 11(b) with only the central R_z controlled, is the sole system-size-dependent element ($\approx 4.2 \times 10^2$ CNOTs per Trotter step at $k = 4$).

the infidelity tracks the exact-evolution curve down to 1.2×10^{-4} at $M = 6$ with acceptance 0.989. The intermediate $\delta t = 0.5$ settles at a floor of 2×10^{-3}. The precise statement behind these floors is the following. With exact evolution on resonance every post-selection operator $K_t = \frac{1}{2}(\mathbb{1} + e^{iE_0t}e^{-iHt})$ leaves the ground-state amplitude unchanged and all K_t share the eigenvectors of H, so the accepted ensemble converges to the ground state and the acceptance to p. With the product-formula evolution $\tilde{U}(t)$ the operators $\tilde{K}_t = \frac{1}{2}(\mathbb{1} + e^{iE_0t}\tilde{U}(t))$ no longer preserve the true ground state, and since the schedules are drawn afresh for every attempt the accepted ensemble after M cycles is $\rho_M = \mathcal{K}^M(\rho_{\mathrm{in}})/\operatorname{Tr}\mathcal{K}^M(\rho_{\mathrm{in}})$ with the averaged map $\mathcal{K}(\rho) = \mathbb{E}_t[\tilde{K}_t\rho\tilde{K}_t^\dagger]$. The long-run state is the dominant eigenoperator of this trace-decreasing map — not the ground state of one effective Hamiltonian, and not a mixture of the separate filters' ground states — and its eigenvalue is the long-run acceptance per cycle. Evaluating $\mathcal{K}$ in the six-dimensional seniority-zero block by quadrature over the half-Gaussian cycle times (notebook, Sec. 19) gives an asymptotic infidelity of 1.2×10^{-4} at $\delta t = 0.25$ with eigenvalue 0.9979, 1.5×10^{-3} at $\delta t = 0.5$ with eigenvalue 0.966, and 0.21 at $\delta t = 1$ with eigenvalue 0.64; these reproduce the floors of Fig. 13 and show that the long-run acceptance of the digital filter decays geometrically, by 0.2% per cycle at $\delta t = 0.25$, rather than tending to p as the ideal theorem states. Coherent product-formula errors thus both shift the located energy and bias the prepared state; the rejection of grossly inaccurate evolutions seen at $\delta t = 1$ is a useful diagnostic, not a guarantee that smaller coherent errors are removed.

C. Gate budget at matched fidelity: which input should the filter get?

The question that decides the value of the refinement stage is not whether the pipeline beats an *unfiltered* variational state — a few rodeo cycles improve any input with large overlap — but whether refinement followed by the filter beats the same filter applied to the best input that is available cheaply, or no filter at all. We therefore feed one and the same Trotterized filter (second order, $\delta t = 0.25$, $\sigma = 4$, $|t_k|$ drawn from a Gaussian, 40 common time sets drawn afresh per attempted preparation) eight inputs: (i) the prolonged $k = 2$ deter-

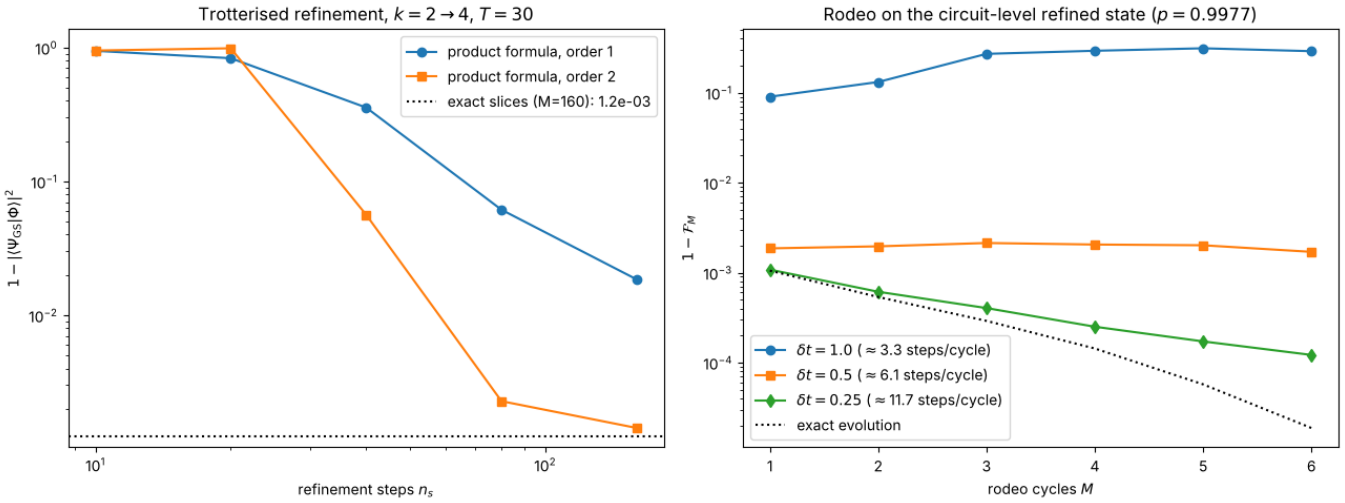


FIG. 13. The pipeline at the circuit level ($k = 4$, $N = 4$, $g = 1$). Left: infidelity of the Trotterized refinement versus the number of product-formula steps n_s along the $T = 30$ path, for first- and second-order formulas; the dotted line is the exact-slice reference. Right: rodeo infidelity versus the number of cycles M for a Trotterized controlled evolution with fixed step size δt (input: the $n_s = 80$ circuit-level refined state, $p = 0.9977$). Too-coarse steps are rejected by the filter itself (the $\delta t = 1$ acceptance collapses to 0.10); $\delta t = 0.25$ tracks the exact evolution down to the floor of the averaged post-selection map, 1.2×10^{-4}.

minant, $p = 0.868$, gate-free; (ii) the four-pair restricted ansatz of Sec. V D, $p = 0.99982$, 192 CNOTs; (iii) the full-template gate-level UCCSD state, the same state for 1 312 CNOTs; (iv) the prolonged $k = 3$ coarse state of Sec. IV C, $p = 0.964$, prepared with two pair rotations (96 CNOTs) and fed to the filter *without* refinement; (v) an ordinary adiabatic ramp from the diagonal Hamiltonian, $H(s) = D + \sin^2(\pi s/2) \sum_{pq} h_{pq}$ started from the HF determinant, which has no shift and no coarse term and whose seniority-zero gap at $s = 0$ is 2.0: it reaches $p = 0.9987$ at $T = 4$ with exact slices and $p = 0.9978$ with 11 second-order steps, $2 \times 11 \times 296 = 6\,512$ CNOTs; (vi) the Trotterized refined state at $T^* = 16$ (43 second-order steps, $p = 0.9922$, $2 \times 43 \times 310 = 26\,660$ CNOTs including the coarse term); (vii) the refined state at $T = 30$ of Fig. 13 (80 steps, $p = 0.9977$, 49 600 CNOTs); and (viii) the $3 \to 4$ chain of Sec. IV C at the counted circuit level ($T = 8$, 22 steps with the coarse term split into its coarse groups, $p = 0.9912$, 28 520 CNOTs). Inputs (iv) and (v) are the two baselines needed to isolate what the specific refinement path contributes: (iv) asks whether refinement is needed after a useful coarse solution, (v) whether the shift-controlled projector path does anything that generic adiabatic preconditioning from the same determinant does not. For each input and each cycle number $M = 0, \ldots, 6$ we track the ensemble infidelity of the accepted runs, the acceptance P_M, the CNOTs of one attempt that runs all M cycles, $C_M^{\text{att}} = C_{\text{prep}} + \sum_{m \le M} \langle c_m \rangle$ — the coherent two-qubit depth proxy of a successful run — and the expected CNOTs per accepted preparation,

$$C_M = \frac{C_{\text{prep}} + \sum_{m=1}^{M} \langle P_{m-1}\, c_m \rangle}{\langle P_M \rangle}, \qquad (12)$$

where $c_m = 2 \times 416\, n_m$ is the cost of the m-th controlled cycle, the expectation is over the random schedules, and the factor P_{m-1} inside it accounts for early termination of a run at the first ancilla outcome 1 (without early termination the numerator is simply $C_{\text{prep}} + \sum_m \langle c_m \rangle$; the difference is below 5% for all inputs). Rejected attempts add to the total work but not to the coherent depth of the accepted state. Equation (12) counts repeated state preparation only. One-time costs are reported separately and are not comparable across routes without a shot model: the UCCSD training took eight L-BFGS iterations in a noiseless simulation, each requiring the energy and its gradient — a hardware shot budget would need an estimator, a grouping scheme, a shot precision and a stopping criterion — and the energy search of the rodeo stage costs the scans of Sec. IV F, of order 10^3 four-cycle runs times their shots, with a preparation cost and peak statistics that depend on which input is scanned. For Q requested accepted states the natural total is $C_{\text{setup}} + Q\, C_M$, and the setup is amortized away for large Q.

Figure 14 and Table III give the result. The first observation is that the high-overlap inputs approach a common product-formula-limited accuracy, $1 - \mathcal{F} \simeq 1.1$–$1.7 \times 10^{-4}$ after six cycles, the floor of the averaged post-selection map at $\delta t = 0.25$ (Sec. V B); the determinant, which still carries substantial excited weight at $M = 6$ (1.8×10^{-3}), and the prolonged $k = 3$ state (7.6×10^{-4}) need more cycles to reach that regime, and six-cycle values are not measured asymptotes. The second observation is the ranking at *matched accuracy*, read off the minimum cost over inputs and cycle numbers, $M = 0$ included, for a target infidelity: for every target above the

floor — 2×10^{-3}, 5×10^{-4} and 2×10^{-4} — the cheapest route is the unfiltered four-pair ansatz, 192 CNOTs at 1.8×10^{-4}, followed by the same state in its 1 312-CNOT template; the cheapest route through the filter is the HF ramp, with 1.7×10^4, 4.0×10^4 and 6.4×10^4 CNOTs per accepted preparation after one, three and five cycles; and the cheapest refinement route, the $T^* = 16$ chain, costs 4.9×10^4, 7.4×10^4 and 9.4×10^4 after two, four and six cycles. Refinement followed by the filter is therefore tied within a factor 1.5 with filtering the unrefined determinant (which reaches 1.8×10^{-3} after six cycles for 7.0×10^4 CNOTs), costs 2.5–3 times more than the plain adiabatic ramp at every target, and costs 200–500 times more than the four-pair preparation, which needs no filter; the prolonged $k = 3$ coarse state filtered directly reaches 7.6×10^{-4} after six cycles for 6.7×10^4 CNOTs, cheaper than refining it first. The cost ratios quoted in an earlier version of this work at a fixed cycle number compared states at different achieved errors and are superseded by these matched-accuracy figures; none of the operating points is a globally optimized minimum, since refinement time, shift and step size were not searched. The claim that the hybrid undercuts the variational preparation by orders of magnitude, which compared the UCCSD *energy error* with the pipeline *infidelity*, does not survive the comparison of like with like.

Table IV records the operating points behind the different fidelities quoted in this paper, which are not contradictory because they belong to different stages of idealization: the ideal filter of Sec. IV F and the digital pipeline of this section differ in refinement time, slice treatment, filter width and evolution, and the residual errors of the former are not achieved by the counted circuits of the latter.

What remains in favor of the refinement stage is qualitative but real. It contains no variational parameters: nothing is trained at full resolution, no expectation values are measured during state preparation, and the only classical inputs are the shift μ, the schedule T and the step count n_s — which, as noted in Sec. III C, were calibrated here against the exact answer rather than from coarse-space information. Its gate count is dominated by the refinement, which scales with the Trotter step of the fine Hamiltonian and with T^*, and is independent of how hard the variational landscape is. Whether this trade — trainability for depth — pays off is a question about regimes in which the full-space variational input is poor or expensive to train. For the pairing model with $N = 4$ that regime is not reached at any size we can validate exactly (Sec. V E), and the ramp baseline adds that, where an adiabatic stage is wanted, the shift-controlled path is not the cheapest one; we state both as the main negative findings of the present assessment.

D. Remarks on adaptive and improved VQE

The steep UCCSD gate count is not irreducible, and much recent work aims precisely at shallower variational circuits (see Ref. [18] for a review). ADAPT-VQE [16] grows the ansatz one operator at a time, at each step appending the pool operator with the largest energy gradient and re-optimizing all amplitudes; it typically reaches a target accuracy with a small fraction of the full UCCSD operator count and correspondingly shallower circuits. The pairing model makes the mechanism transparent: because the Hamiltonian conserves seniority and the reference is seniority-zero, the gradients of all singles and of all pair-breaking doubles vanish identically on the seniority-zero manifold, so a selection by gradient would start from the $\frac{N}{2}(k - \frac{N}{2})$ pair-hopping doubles. We have *demonstrated* the corresponding restricted ansatz (Sec. V C): the four occupied-pair→virtual-pair rotations of the pool, in the order of the full ansatz, optimized on their own, give $E = 0.636987$ and infidelity 1.78×10^{-4} — the full 26-operator UCCSD result to all displayed digits, whose other 22 amplitudes are exactly zero at the optimum — for $4 \times 48 = 192$ CNOTs in place of 1 312. This is an explicit restricted variational calculation, not an ADAPT run: the symmetry selection rule says which gradients vanish, not that ADAPT selects exactly four operators, each once, nor what error it reaches, and it says nothing about unconstrained optima of an ordered ansatz. Qubit-ADAPT-VQE [17] pushes further by drawing the pool from individual Pauli strings, removing the eightfold string overhead per fermionic double. These developments are complementary to the present pipeline rather than competing with it: the filter is agnostic to how its input is produced, so an ADAPT-style state — at the coarse or at the full resolution — slots in directly. An ADAPT-VQE comparator proper remains the natural next step of the assessment. Conversely, no variational compression by itself supplies the geometric, optimizer-free convergence that the rodeo filter provides once the overlap is large. A different route to a shallower circuit — keeping the Trotterized refinement path as the ansatz structure and optimizing a few of its layers, as in the optimized nucleation of Ref. [9] — is taken up in Sec. VI.

E. Other model-space sizes, particle numbers, and observables

To see whether the ranking of Table III is an accident of the smallest nontrivial case, Table V repeats the comparison, with exact-slice refinement and exact controlled evolution ($\sigma = 3$, 40 time sets), for $N = 4$ and $N = 6$ particles on $k = 4$ and $k = 6$ levels (eight and twelve qubits), always starting from the filled determinant of the smallest coarse space ($k_{\mathrm{low}} = N/2$). The pattern is the same throughout: the prolonged overlap falls to 0.78–0.80 at $k = 6$, refinement for $T = 30$ re-

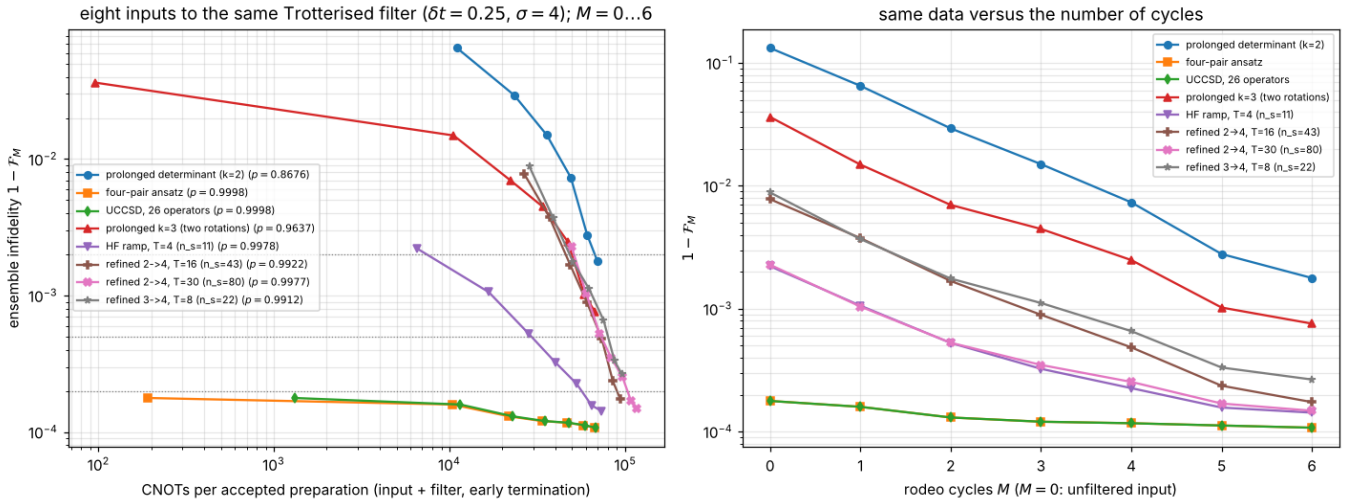


FIG. 14. Matched-observable comparison of eight inputs to the same Trotterized rodeo filter ($k = 4$, $N = 4$, $g = 1$; second order, $\delta t = 0.25$, $\sigma = 4$, 40 common Gaussian time sets): the prolonged $k = 2$ determinant, the four-pair restricted ansatz and the full UCCSD template (the same state at 192 and 1 312 CNOTs), the prolonged $k = 3$ coarse state, the HF adiabatic ramp, the Trotterized resolution-refined states at $T^* = 16$ and $T = 30$, and the $3 \to 4$ chain at $T = 8$ (refinement costs include the coarse term of Eq. (4)). Left: acceptance-weighted ensemble infidelity of the accepted runs versus CNOTs per accepted preparation, Eq. (12), as the number of cycles grows from $M = 0$ (unfiltered input; omitted for the gate-free determinant) to $M = 6$; dotted lines mark the targets 2×10^{-3}, 5×10^{-4} and 2×10^{-4} of the text. Right: the same infidelities versus M. The high-overlap inputs converge to the floor of the product-formula filter; the four-pair ansatz is below every target before any cycle.

TABLE III. Gate budget at $k = 4$ ($N = 4$, $g = 1$, eight system qubits) for eight inputs to the same Trotterized rodeo filter ($\delta t = 0.25$, $\sigma = 4$, 40 time sets). p is the input overlap, $C_{\rm prep}$ the CNOT count of the input (refinement: $2\,n_s$ second-order half-steps of $296 + 14$ CNOTs for the $2 \to 4$ chains and of $296 + 350$ for the $3 \to 4$ chain, plus the coarse preparation; ramp: $2n_s \times 296$), $1 - \mathcal{F}_M$ the acceptance-weighted infidelity of the accepted runs, P_M the acceptance, $C_M^{\rm att}$ the CNOTs of one attempt running all M cycles and C_M the CNOTs per accepted preparation of Eq. (12). One-time training (variational inputs) and the energy scan are not included (see text). The UCCSD energy error of 1.4×10^{-3} (Table I) corresponds to the infidelity 1.8×10^{-4} listed here.

			$M = 0$	$M = 2$				$M = 6$			
input to the filter	p	$C_{\rm prep}$	$1 - \mathcal{F}_0$	$1 - \mathcal{F}_2$	P_2	$C_2^{\rm att}$	C_2	$1 - \mathcal{F}_6$	P_6	$C_6^{\rm att}$	C_6
prolonged determinant (k=2)	0.868	0	1.3×10^{-1}	2.9×10^{-2}	0.887	2.2×10^4	2.3×10^4	1.8×10^{-3}	0.854	6.6×10^4	7.0×10^4
four-pair ansatz (4 parameters)	0.99982	192	1.8×10^{-4}	1.3×10^{-4}	0.995	2.2×10^4	2.2×10^4	1.1×10^{-4}	0.986	6.6×10^4	6.7×10^4
UCCSD template (26 parameters)	0.99982	1 312	1.8×10^{-4}	1.3×10^{-4}	0.995	2.3×10^4	2.3×10^4	1.1×10^{-4}	0.986	6.7×10^4	6.8×10^4
prolonged k=3 state (96-CNOT prep.)	0.964	96	3.6×10^{-2}	7.0×10^{-3}	0.964	2.2×10^4	2.2×10^4	7.6×10^{-4}	0.948	6.6×10^4	6.7×10^4
HF adiabatic ramp, T=4 (n_s=11)	0.9978	6 512	2.2×10^{-3}	5.3×10^{-4}	0.994	2.8×10^4	2.8×10^4	1.4×10^{-4}	0.985	7.3×10^4	7.3×10^4
refined $2 \to 4$, T^*=16 (n_s=43)	0.9922	26 660	7.8×10^{-3}	1.7×10^{-3}	0.989	4.8×10^4	4.9×10^4	1.7×10^{-4}	0.979	9.3×10^4	9.4×10^4
refined $2 \to 4$, T=30 (n_s=80)	0.9977	49 600	2.3×10^{-3}	5.3×10^{-4}	0.994	7.1×10^4	7.2×10^4	1.5×10^{-4}	0.985	1.2×10^5	1.2×10^5
refined $3 \to 4$, T=8 (n_s=22)	0.9912	28 520	8.8×10^{-3}	1.8×10^{-3}	0.988	5.0×10^4	5.1×10^4	2.7×10^{-4}	0.978	9.5×10^4	9.6×10^4

stores it to 0.997–0.999, and the UCCSD input, whose infidelity grows from 1.8×10^{-4} to 1.0×10^{-3} over the table, still enters the filter with a higher overlap than the refined state and leaves it with a lower infidelity at every cycle number. (At $N = 6$, $k = 4$ the ansatz is exact: three pairs on four levels are the particle–hole conjugate of one pair, for which UCCSD is complete.) The price of the UCCSD input is its circuit — 6 976 CNOTs and 92 parameters at $k = 6$, $N = 4$; 9 024 CNOTs and 117 parameters at $k = 6$, $N = 6$ — and the training of those parameters, which the refinement avoids at the cost of $2n_s(N_{\rm CNOT}^{\rm step} + N_{\rm CNOT}^{A})$ CNOTs with $N_{\rm CNOT}^{\rm step} = 732$ at $k = 6$. This is a controlled small-system demonstration, not a scaling study; a regime in which refinement wins would require a Hamiltonian for which the variational input is poor, and we do not find one within the pairing model at these sizes. The first step beyond the pairing force — opening the one-particle–one-hole channel, which switches on the UCCSD singles and degrades the single-reference ansatz with the strength of the pair breaking — is taken in the companion paper [10].

a. Observables. A state-preparation method for nuclear structure is judged by the observables of the state it delivers, not only by its overlap. Table VI lists the level

TABLE IV. Operating points behind the fidelities quoted in this paper ($k = 4$, $N = 4$, $g = 1$).

result	assumptions and operating point	status
coarse state (Sec. III A)	factorized UCCSD, numerically optimized	valid circuit family; $\epsilon_{\rm low}$ to be supplied or bounded
prolongation (Sec. IV B)	nested bare Hamiltonians, empty appended levels	gate-free embedding; expectation-value identity exact
adiabatic guarantee, Eq. (6)	exact coarse eigenstate, isolated branch, ideal continuous evolution	conditional bound, not the complete digital budget
refinement (Sec. IV B)	$T = 25$ or 30, exact midpoint slices (160)	ideal; slice error $\sim 10^{-6}$
filter (Sec. IV F)	$T = 25$ refined input, $\sigma = 3$, exact controlled evolution, 40 schedules	ideal; sampling error up to 80% at $M = 10$
circuit comparison (Secs. V B–V C)	product formulas, $T = 16$ or 30, $\sigma = 4$, step ceiling $\delta t = 0.25$	digital; floor 1.2×10^{-4}, acceptance decaying 0.2% per cycle

TABLE V. The pipeline and the UCCSD comparator at other sizes ($g = 1$): coarse space the filled determinant at $k_{\rm low} = N/2$, exact-slice refinement for $T = 30$, exact controlled evolution with $\sigma = 3$, 40 time sets. ov_0 and p are the overlaps of the prolonged and refined states with the exact ground state, the UCCSD columns give the infidelity and energy error of the sector-level UCCSD state together with its parameter and CNOT counts, and the last four columns the acceptance-weighted infidelity after M rodeo cycles from the refined and from the UCCSD input. The register has $2k$ qubits (8 and 12) and the uncontrolled first-order Trotter step of Eq. (11) costs 296 CNOTs at $k = 4$ and 732 at $k = 6$.

(k, N)	dim	gap	ov_0	$p\,(T{=}30)$	$1 - \mathcal{F}_{\rm UCCSD}$	$\Delta E_{\rm UCCSD}$	params	CNOTs	refined+rodeo		UCCSD+rodeo	
									$M{=}2$	$M{=}6$	$M{=}2$	$M{=}6$
$(4, 4)$	70	1.823	0.868	0.9988	1.8×10^{-4}	1.4×10^{-3}	26	1 312	3.4×10^{-4}	2.3×10^{-5}	4.1×10^{-5}	1.4×10^{-6}
$(6, 4)$	495	2.017	0.798	0.9976	7.6×10^{-4}	6.1×10^{-3}	92	6 976	6.3×10^{-4}	4.1×10^{-5}	1.7×10^{-4}	7.6×10^{-6}
$(4, 6)$	28	1.779	0.894	0.9991	$< 10^{-15}$	$< 10^{-15}$	15	688	2.5×10^{-4}	1.7×10^{-5}	0	0
$(6, 6)$	924	2.052	0.780	0.9972	1.0×10^{-3}	8.3×10^{-3}	117	9 024	7.2×10^{-4}	4.8×10^{-5}	2.4×10^{-4}	1.0×10^{-5}

occupation numbers $n_p = \sum_\sigma \langle n_{p\sigma} \rangle$ and the correlation energy $E - E_{\rm ref}$ of the states of Table III at $k = 4$, $N = 4$, $g = 1$ (for the filtered states, the acceptance-weighted average over the 40 time sets). The exact ground state depletes the two occupied levels by 0.07 and 0.20 particles. The UCCSD state reproduces the occupations to 3×10^{-3} and the correlation energy to 1.4×10^{-3}; the refined state at $T = 30$, whose infidelity is 2.3×10^{-3}, deviates by up to 0.02 in the occupations and by 10^{-2} in the correlation energy, and two rodeo cycles reduce both deviations by a factor 2–3, six cycles by a factor 4–5. Occupation numbers converge more slowly than the energy, as expected for a quantity that is linear rather than quadratic in the error of the state.

TABLE VI. Level occupation numbers n_p and energy of the prepared states at $k = 4$, $N = 4$, $g = 1$ ($E_{\rm ref} = 1$); the filtered states use the Trotterized filter of Table III ($\delta t = 0.25$) and are acceptance-weighted averages over 40 time sets; $a\,10^{-b}$ stands for $a \times 10^{-b}$.

state	n_0	n_1	n_2	n_3	$E - E_{\rm ref}$	$1 - \mathcal{F}$
exact	1.929	1.803	0.197	0.071	−0.3645	0
reference det.	2	2	0	0	0	$1.3\,10^{-1}$
UCCSD	1.930	1.806	0.194	0.070	−0.3630	$1.8\,10^{-4}$
refined ($T{=}30$)	1.908	1.795	0.205	0.092	−0.3549	$2.3\,10^{-3}$
refined+rodeo $M{=}2$	1.920	1.798	0.202	0.080	−0.3621	$5.3\,10^{-4}$
refined+rodeo $M{=}6$	1.923	1.801	0.199	0.077	−0.3637	$1.5\,10^{-4}$
UCCSD+rodeo $M{=}2$	1.924	1.801	0.199	0.076	−0.3637	$1.3\,10^{-4}$

VI. DISCUSSION AND OUTLOOK

The pairing model lets us exhibit the complete refinement-plus-filter chain with every stage benchmarked against exact results: a gate-level UCCSD coarse solver, a gate-free prolongation with a counted conditioned coarse term, a gapped adiabatic refinement whose extracted time responds to the shift-controlled path gap and to the coupling without identifying a scaling law, and a rodeo filter whose ensemble infidelity follows Eq. (10) — all validated a second time at the circuit level with product formulas. The two algorithms are complementary in the intended sense: refinement raises the overlap from 0.87 to 0.998, and the filter removes the adiabatic tail geometrically. The coupling scan adds a practical lesson: at strong pairing the limiter is not the path gap but the quality of the prolonged starting state. The register and the circuits of all stages — prolongation, the adiabatic path, the shift rule and the rodeo cycle — are written for the generic $2k$-qubit register and do not exploit seniority for compression; the favorable dynamics and the small variational inputs available here do rely on the seniority conservation of the pairing force, and the companion paper [10] examines what changes when a particle-hole

interaction opens the one-particle–one-hole channel, the price of the more realistic interaction appearing in the Pauli-string content of the Hamiltonian and hence in the depth of each Trotter step.

The central quantitative finding, however, is negative for the refinement stage as a gate-saving device. Compared on the same observable — the ground-state infidelity — and with the same Trotterized filter, refinement followed by the rodeo algorithm is tied with filtering the unrefined prolonged determinant, costs four times more than an ordinary adiabatic ramp from the Hartree–Fock Hamiltonian for the same output overlap, and is beaten by two orders of magnitude at every target accuracy by the four-parameter pair-rotation ansatz, which needs no filter at all, at every size we can validate exactly (Tables III and V). The reason is simple, and Sec. V C states it as the ideal large-gap law $1 - \mathcal{F}_M \simeq \frac{1-p}{p} 2^{-M}$: a better input saves cycles only logarithmically in its odds ratio, so moving from $p = 0.868$ to 0.998 saves some six cycles and raises the asymptotic acceptance from 0.87 to 1.00, and whether those savings repay tens of thousands of refinement gates is a separate question that, in these systems, they do not — a single-reference variational circuit with four parameters already has an overlap above 0.9998, and an overlap of 0.87 costs the filter only a few extra cycles. What the refinement offers in exchange is that it is optimizer-free and measurement-free during preparation, and that its cost is set by the Hamiltonian evolution rather than by the trainability of a landscape. Whether that trade pays off is a question about regimes we cannot reach with exact validation — larger spaces, interactions for which single-reference ansätze fail, or hardware on which the training measurements of a variational loop dominate — and we regard the present work as the controlled baseline against which such claims should be measured, not as evidence for them.

a. Relation to symmetry-restoration and subspace approaches. The results invite a comparison with the pairing-model work of Lacroix and collaborators summarized in the Introduction. Both programmes start from a cheap, imperfect trial state and sharpen it; the difference is what is sharpened and how. In the symmetry-restoration route the trial state breaks a symmetry on purpose and the projector — phase estimation on the symmetry operator [23], an oracle [28] or a measurement-side projection [31] — removes the unwanted sectors, after which a Krylov or phase-estimation stage refines the energy [26]. Our coarse state conserves particle number and spin by construction; what it lacks is *resolution*, and the refinement supplies it along a gapped path while the rodeo cycle filters in *energy* rather than in a symmetry quantum number. The two filters are structurally the same resource — one ancilla, a controlled unitary and a post-selected measurement — and they compose: a symmetry-projected input would enter our pipeline exactly as the UCCSD state does. The quantum subspace methods of Refs. [30, 32, 33] are complementary in a different way: they trade coherent depth for measurements and a classical generalized eigenvalue problem, and their accuracy is bounded by the quality of the states that span the subspace, whereas the rodeo filter converges geometrically in depth but requires the controlled evolution. Two observations of that work bear on ours, with the limits of an energy filter kept in mind. Ref. [44] documents that ADAPT-VQE with a symmetry-breaking pool can collapse to unphysical energies on degenerate pairing systems. A downstream energy filter can repair such an input only insofar as the input retains a nonzero overlap with the target: it cannot recover a state of zero overlap, cannot move weight between particle-number sectors that the evolution conserves, and cannot separate exactly degenerate states by energy alone, so a symmetry-broken ADAPT state would have to be symmetry-projected before, not instead of, the filter. Ref. [33] shows that the accuracy of the one-body Green's function of a small superfluid system, computed with the pure pairing Hamiltonian across the normal-to-superfluid transition driven by the pairing strength, is controlled by the quality of the even-N ground state; a filtered state of the kind prepared here is a possible input to that construction. The compact pair-to-qubit encoding used there [23, 33] is available precisely because the pairing force conserves seniority; the particle-hole term of the companion paper [10], which is not part of the model of Ref. [33], is the ingredient that forces one out of it and onto the full $2k$-qubit register used throughout this work.

b. Relation to nucleation. The nucleation method of Ref. [9] is the closest relative of the refinement stage, and the comparison is instructive in both directions. Both start from a small problem solved exactly on a classical computer and compiled into a circuit, and both reach the target through a sequence of Trotterized adiabatic interpolations with the $\sin^2(\pi s/2)$ schedule of Eq. (3), each of which changes the Hamiltonian only locally: nucleation switches on the couplings of newly added lattice sites, initialized in $|+\rangle$, to the existing bulk; refinement switches off the conditioned coarse term A of Eq. (4) and switches on the matrix elements of H_{high} that connect the occupied coarse levels to the appended ones, initialized empty in $|0\rangle$. Nucleation grows the domain at fixed resolution, refinement the resolution at fixed domain, and the two compose in principle: a lattice or shell-model problem could be nucleated to its physical size in a coarse basis and then refined to the full basis, or the reverse, with the same Trotterized machinery and the same filter at the end.

What Ref. [9] adds to the present assessment is twofold. First, it provides the rigorous sufficient conditions on the total time and on the Trotter step count of each stage quoted in Sec. III C, and shows on a twelve-site transverse-field Ising chain that they overestimate the resources actually needed by orders of magnitude: an overlap of 0.997 is reached with $T \simeq 40$ and some 2×10^3 Trotter steps in total, against bounds of $T \geq 2.6 \times 10^3$ and $N \geq 2 \times 10^{10}$. Section IV E makes the same compar-

ison for the refinement path: the time bound of Eq. (6) overshoots the empirical T^* by a factor 3×10^2 and the step bound overshoots the steps actually needed by a factor 2×10^2 (Table II), because the bound charges the full norm of $B - A$ where the dynamics only pays for the matrix elements that couple the tracked state to its neighbors; the distance between bound and practice is the room in which the inverse-gap scaling of Ref. [7] operates. Second, and more directly relevant to our negative finding, Ref. [9] introduces *optimized nucleation*: the Trotterized growth circuit is kept as the ansatz structure and its Trotter angles as the initial parameters, and a small number L of layers per stage — far below the Trotter step count of the non-variational version — is then optimized layer by layer to minimize the energy. On a 5×5 transverse-field Ising model this reaches a lower energy error at fewer two-qubit gates than converting a matrix-product state to a circuit, even when the latter is itself re-optimized. The same construction is available for refinement: the n_s slices of Eq. (3) compiled in Sec. V B form a parameterized circuit whose angles are fixed by $(s, \delta t)$, and an *optimized refinement* with a few layers per stage would trade the one asset the adiabatic path has over the UCCSD input in Table III — freedom from classical optimization and training measurements — for a gate count that could fall well below the 4.96×10^4 CNOTs of the $T = 30$ path. Whether a warm-started, physically structured ansatz of this kind escapes the trainability problems of a generic one is the question Ref. [9] raises for lattices; the pairing model, whose optimum is known exactly at every resolution, is the place to settle it, and we list it among the extensions below.

c. Hardware requirements. The Trotterized pipeline of Sec. V B quantifies the dominant coherent error of an ideal device, and the requirements it implies are explicit: $2k+1$ qubits (9 at $k = 4$, 13 at $k = 6$, 17 at $k = 8$); a mid-circuit measurement with reset and reuse of one ancilla; arbitrary-angle R_z rotations, 160 of them per UCCSD preparation and 60 per Trotter step; and a coherent budget that ranges from 192 two-qubit gates for the cheapest route of Table III (the four-pair preparation, no filter) through 1.7×10^4 per accepted preparation for the HF ramp with one cycle to 10^5 for the refined input with six cycles (the companion paper [10] gives the corresponding budget with the particle-hole term), on top of the energy scan and the repetitions needed to estimate acceptance probabilities. At the two-qubit error rates demonstrated in large-scale experiments today — of order 10^{-2} on the superconducting processor of Ref. [55] and $(0.8\text{–}2) \times 10^{-3}$ on trapped-ion processors [56, 57] — a 10^4-gate run suffers of order ten faults even on the best of these and a 10^5-gate run of order one hundred; none of these budgets is executable with physical qubits without substantial error mitigation, and qubit counts alone say nothing about a device's suitability. Early logical-qubit demonstrations on neutral-atom arrays have executed circuits on tens of logical qubits with error detection and, in part, error correction and mid-circuit readout [58, 59], and the rodeo algorithm has been run in two-qubit form on trapped-ion and superconducting hardware [6, 47]; a logical-depth implementation of the budgets above remains to be demonstrated. A proportionate near-term target is therefore a mitigated $k = 3$ demonstration: the $2 \to 3$ chain of Sec. IV B on seven qubits (six system qubits plus the ancilla), whose $N = 4$ coarse state is a single determinant, or the $3 \to 4$ chain of Sec. IV C on nine qubits, with a short refinement schedule and one or two rodeo cycles at coarse δt, i.e. 10^3–10^4 two-qubit gates, where the collapse of the acceptance under a corrupted evolution (Sec. V B) doubles as a hardware diagnostic. All-to-all connectivity absorbs the Jordan–Wigner ladders without SWAP overhead; on square or heavy-hex lattices the weight-4 pair-hopping rotations act on adjacent qubits, which limits but does not eliminate routing costs. The full $k = 4$ budget is a workload for the first generation of logical-qubit machines such as the neutral-atom system planned by the QuNorth initiative [54], and Table III is offered as its specification.

d. Concluding remarks and perspectives. The present results establish, on a fully controlled example, how resolution refinement followed by the rodeo algorithm behaves at the circuit level — every stage compiled, counted and validated against exact results — and what it is worth: it prepares the correlated ground state to the accuracy of the Trotterized evolution without any variational optimization, at a gate cost that, for the pairing model at these sizes, exceeds that of filtering a cheap variational input. Three extensions would sharpen the assessment. First, comparators and compilation: an ADAPT-VQE run proper, a pair-encoded (one qubit per level) variational and filtering baseline and a symmetry-projected BCS-like input, an optimized refinement in the sense of Ref. [9] — the Trotterized path of Eq. (3) as a warm-started variational circuit with a few layers per stage — and a compilation effort applied equally to every route (merged half-steps, optimized excitation circuits, routing on a concrete connectivity), the minimum-cost search of Table III being then repeated over refinement time, shift and step size with the shot budgets of training and energy search amortized over the number of states requested. Second, Hamiltonians for which single-reference variational inputs fail — shell-model interactions with competing deformations, or the strong-coupling regime of Sec. IV D treated with a nontrivial coarse space — which is where the trainability argument for refinement would be tested rather than asserted; the companion paper [10] takes the first step in this direction by adding a seniority-breaking particle-hole interaction, which degrades the single-reference ansatz with the strength of the pair breaking. Third, hardware: a mitigated seven- or nine-qubit demonstration on present devices, where the acceptance statistics of the filter double as a diagnostic of the compiled evolution, followed by the full $k = 4$ budget on an early logical-qubit machine.

The pairing model, exactly solvable at every resolution, remains the calibration standard for all three.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian, gate-level UCCSD–VQE, prolongation, adiabatic refinement, and the ancilla-based rodeo algorithm — is provided as a tested Python package and a single Jupyter notebook that reproduces every figure in this article and in the companion paper [10]. Give GitHub link when everything is ready.

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